# Cluster-Adaptive Prompting for Multi-Turn Travel and Tourism Conversational Systems

## Final Report

Surnella Sandeep

Department of Data Analytics, Walsh College

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Mr. Smitraj Raut

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**GitHub repository:** https://github.com/surnella/QM640-Dialogue-Clustering-Capstone

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## Abstract

Online travel agencies increasingly deploy large language model (LLM) assistants for vacation-package negotiation, yet one-size-fits-all prompts ignore customer decision-making heterogeneity while unguarded few-shot prompting risks entity hallucinations and policy violations. This study proposes and evaluates an integrated neuro-symbolic framework for archetype-aware, policy-governed, and turn-efficient conversational artificial intelligence (AI) in travel booking. Evaluated on 1,334 Microsoft Frames dialogues (70% golden discovery, $n = 933$; 30% held-out test, $n = 401$) and 1,499 MultiWOZ 2.2 comparison dialogues, the research follows a four-stage progression: **Identify (RQ1) → Personalize (RQ2) → Govern (RQ3) → Optimize (RQ4)**. First, to **identify** customer interaction styles early (**RQ1**), clustering algorithms differed significantly across 159 runs (53 matched $n = 500$ subsamples), one-way $F(2, 156) = 91.33$, $p < .001$, $\eta^2 = .54$ (matched repeated-measures $F(1.72, 89.30) = 83.04$, Friedman $\chi^2(2) = 56.30$, $p < .001$). Selecting *K*-Means ($k = 3$) for 100% routing coverage (versus HDBSCAN's 39.1% MiniLM / 35.8% Gemini noise rejection) on a 46-dimensional hybrid *Blocks* representation over the first nine user messages established three stable reference archetypes (bootstrap ARI = .89 in the truncation grid; .87 in the 6-configuration ablation): *Decisive Booker*, *Open Explorer*, and *Rigid-Constraint Seeker*. Prefix Logistic Regression routers achieved Cohen's $\kappa = .535$ (70.1% agreement) after five user messages and $\kappa = .958$ (97.3% same-feature consistency) after nine messages against the nine-message reference labels. Second, to **personalize** assistant communication (**RQ2**, $N = 33$ repeated-measures dialogues), routing via the five-message *Blocks* router into a combined package of archetype-partitioned FAISS sub-indices and pacing directives improved de-lexicalized SBERT style similarity from .453 (static few-shot) and .530 (zero-shot) to .597, $F(1.88, 60.29) = 9.79$, $p < .001$, $\eta_p^2 = .23$, and cut pacing error from 10.12

words (zero-shot) and 6.33 words (static) to 3.87 words, $F = 7.10$, $p = .007$, whereas generic semantic metrics stayed invariant and coarse whole-dialogue routing ($\kappa = .24$) had yielded null effects ($p = .211$). Third, to **govern** business and factual correctness (**RQ3**, 2 × 2 factorial, $N = 144$ evaluations across 36 dialogues; dual-judge $\kappa = .917$), pairing cluster-adaptive prompting with a declarative rule engine, [SLOT] masking, and leakage sanitization raised violation enforcement sensitivity from 5.6% to 97.2% ($F(1, 68) = 498.78$, $p < .001$; compliant specificity = 88.9% vs. 94.4%, $p = .408$; overall rule accuracy = .931, $F(1, 140) = 43.92$, $p < .001$), boosted Slot $F_1$ to .952 ($F = 489.52$, $p < .001$), and eliminated the .359 (55.6% incidence) unguarded exemplar leakage rate to .000 (interaction $F(1, 140) = 34.82$, $p < .001$; Welch $t(35) = 5.90$, $p < .001$) while retaining .506 style similarity. Fourth, to **optimize** multi-turn booking efficiency (**RQ4**, 2 × 3 simulation, $N = 180$ unique seed dialogues, $n = 30$/cell; persona $\kappa = .933$), a *Verified Informed Slot Ledger* and archetype-specific policies reduced total turns (user plus assistant messages) to package agreement by 42.7% for *Rigid-Constraint Seekers* (8.73 → 5.00, reaching the 5-turn protocol floor), 34.2% for *Decisive Bookers* (7.60 → 5.00), and 29.5% for *Open Explorers* (7.47 → 5.27; $F(1, 174) = 664.43$, interaction $F(2, 174) = 10.89$, both $p < .001$) while eliminating redundant slot re-asking (.109 → .000). Combining behavioral routing, adaptive communication, explicit policy controls, and state-aware dialogue management thus supports personalized, compliant, and efficient travel-booking interactions.

*Keywords:* conversational artificial intelligence, customer archetypes, large language models, few-shot prompting, business-rule enforcement, slot filling, dialogue management, travel booking

# Introduction

## Background and Context

Conversational artificial intelligence (AI) assistants have become a core digital channel in travel, tourism, and hospitality, supporting destination discovery, package comparison, and itinerary booking (Buhalis & Sinarta, 2019; Ivanov & Webster, 2019; Jannach et al., 2021). Unlike single-turn question answering, vacation package booking is a consultative, multi-turn negotiation in which travelers balance competing constraints across origin, destination, dates, duration, hotel tier, and budget (El Asri et al., 2017). Empirical analyses of human travel dialogues demonstrate that travelers converse in distinct behavioral styles: some state fixed parameters up front and seek immediate confirmation, others explore multiple destinations and hotel tiers side by side, and others insist on rigid date and luxury constraints that require proactive trade-off resolution when preferred inventory exceeds their budget ceiling (Inaba et al., 2022; Melián-González et al., 2021).

Despite this behavioral diversity, commercial large language model (LLM) travel assistants typically use one static system prompt and a uniform sequential intake script. Asking clarifying questions one slot at a time or offering a single hotel option per turn creates repetitive turns and session abandonment. Conversely, adapting an LLM's style purely through in-context few-shot examples introduces two severe reliability hazards (Liu et al., 2022; Rebedea et al., 2023): first, unmasked retrieved examples cause factual entity leakage, where the model copies stale hotel names, airfares, or promo codes from the examples into the active session; second, stylistic prompt instructions compete with business rules, causing the assistant to overlook mandatory visa windows, passport validity thresholds, party-size limits, or discount caps (Yao et

al., 2024). Resolving both personalization and compliance requires a neuro-symbolic architecture that decouples behavioral style adaptation from deterministic policy verification.

**Problem Statement**

*Business problem statement.* Online travel agencies (OTAs), tour operators, and hospitality contact centers suffer from low self-service conversion and elevated session abandonment because static conversational assistants fail to adapt their information density and recommendation strategy to distinct customer decision-making styles, while unverified generative responses expose enterprises to financial and regulatory liability from booking policy violations and hallucinated package terms (Buhalis & Sinarta, 2019; Ivanov & Webster, 2019; Melián-González et al., 2021).

*Technical problem statement.* Existing LLM dialogue systems lack (a) an empirically validated representation for clustering multi-turn user behavior into stable archetypes that can be routed accurately from early dialogue prefixes without end-of-dialogue outcome leakage, (b) a slot-masked exemplar retrieval mechanism that achieves target archetype style convergence without inducing entity leakage, and (c) formal factorial evidence quantifying the main and interaction effects of external declarative rule verification and cluster-adaptive prompting on slot fidelity, policy adherence, and multi-turn turn economy (Acikgoz et al., 2025; Liu et al., 2022; Rebedea et al., 2023).

*Measurable research formulation.* This study models six quantitative dependent variables ($Y$)—cluster separation Silhouette ($Y_1$), style convergence cosine and pacing error ($Y_2$), policy rule adherence rate ($Y_3$), exemplar entity leakage rate ($Y_4$), factual slot $F_1$ ($Y_5$), and log-transformed turns to goal completion and redundant elicitation rate ($Y_6$)—as functions of four manipulated independent variables ($X$): clustering algorithm and trajectory representation ($X_1$,

RQ1), in-context conditioning strategy ($X_2$, RQ2), rule-engine mode crossed with prompting strategy ($X_3$, RQ3), and prompting strategy crossed with refined user archetype ($X_4$, RQ4).

**Purpose of the Study and Research Questions**

The purpose of this quantitative study is to develop and evaluate an end-to-end neuro-symbolic, cluster-adaptive conversational framework on public human travel dialogues across four linked research questions (Table 1).

**Table 1**

*Research Questions, Hypotheses, and Analytical Framework*

| RQ | Research question | Null and alternative hypotheses | Experimental design and statistical test |
|---|---|---|---|
| RQ1 | Which clustering algorithm (K-Means, Ward agglomerative, HDBSCAN) best separates dialogue embeddings, and which stable, early-routable user archetypes emerge? | $H_{01}$: Equal mean Silhouette across algorithms. $H_{a1}$: At least one differs, and truncated behavioral-semantic prefixes yield stable (ARI ≥ .60), routable archetypes. | Resampling benchmark across $N$ = 159 runs (53 matched $n$ = 500 subsamples: one-way ANOVA, matched RM-ANOVA, Friedman, Welch, Kruskal–Wallis, Tukey & Games–Howell) + truncation grid |
| RQ2 | Does cluster-adaptive few-shot prompting increase style convergence and pacing calibration relative to zero-shot and static few-shot prompting? | $H_{02}$: Equal mean style cosine across conditions. $H_{a2}$: Cluster-adaptive > static and zero-shot on refined user archetypes. | One-way repeated-measures ANOVA ($N$ = 33 dialogues × 3 conditions = 99 calls; Greenhouse–Geisser, Friedman, Holm contrasts) |
| RQ3 | How do a declarative rule engine and cluster-adaptive prompting interact to govern rule adherence, exemplar leakage, slot $F_1$, and style alignment? | $H_{03}$: No main or interaction effects. $H_{a3}$: Active rules raise adherence and slot $F_1$ and eliminate few-shot entity leakage. | 2 × 2 factorial ANOVA ($N$ = 144 across 36 held-out dialogues, $n$ = 36/cell; OLS, HC3, ART, sensitivity/specificity split; dual-LLM judge κ) |
| RQ4 | Does cluster-adaptive prompting with an informed-slot ledger reduce turns to goal completion and redundant elicitation differently across user archetypes? | $H_{04}$: No prompting or prompting × archetype interaction effects. $H_{a4}$: Cluster-adaptive prompting reduces log turns and RER with an archetype interaction. | 2 × 3 between-subjects ANOVA on log turns and RER ($N$ = 180 unique held-out dialogues, $n$ = 30/cell; OLS, HC3, ART, Welch $t$, Mann–Whitney $U$; persona κ) |

*Note.* RQ = research question; ARI = adjusted Rand index; HC3 = heteroscedasticity-consistent SE; ART = aligned rank transform; RER = redundant elicitation rate. All tests use α = .05.

### Contributions and Expected Value

This study delivers four contributions: (1) demonstrating that truncating dialogues to the first nine user messages and combining Gemini prefix embeddings with structured LLM behavioral ratings (*Blocks*) yields three stable reference archetypes (bootstrap ARI = .89 in the truncation grid; .87 in the 6-configuration ablation) with chance-corrected held-out prefix routing agreement of $\kappa = .535$ (70.1% raw agreement) at five user messages and $\kappa = .958$ (97.3% same-feature boundary consistency) at nine user messages against the unsupervised nine-message *K*-Means reference clusters; (2) contrasting primary versus refined archetypes in RQ2 to demonstrate that combining archetype-partitioned exemplar retrieval with routed archetype pacing directives significantly improves de-lexicalized SBERT structural style similarity ($p < .001$) and archetype pacing calibration ($p = .007$) once upstream routing is coherent, while generic semantic similarity remains unchanged; (3) demonstrating in RQ3 that an integrated governance layer (declarative rule engine, [SLOT] masking, and post-generation sanitization) eliminates the 35.9% entity leakage (55.6% incidence) of unguarded few-shot prompting in the evaluated sample while raising policy violation enforcement sensitivity from 5.6% to 98.6% and preserving 88.9% compliant-case acceptance specificity; and (4) showing across 180 independent multi-turn simulations in RQ4 that coupling the *Verified Informed Slot Ledger* with archetype-specific turn-economy policies reduces total conversational turns to booking agreement by 29.5% to 42.7% and eliminates redundant slot questions.

## Literature Review

### Literature Review Approach and Relevance Matrix

Peer-reviewed studies were retrieved from the ACL Anthology, ACM Digital Library, IEEE Xplore, ScienceDirect, Google Scholar, and arXiv using Boolean combinations of *travel*

*and hospitality chatbots*, *conversational recommender systems*, *task-oriented dialogue corpora*, *trajectory clustering*, *in-context exemplar retrieval*, and *LLM guardrails*. Twenty-six sources spanning business adoption, dialogue datasets, unsupervised clustering, retrieval-augmented prompting, and statistical methodology were synthesized. Table 2 maps the core literature streams to the gaps addressed in RQ1–RQ4.

**Table 2**

*Literature Relevance Matrix Across Business and Technical Domains*

| Study and stream | Methodology and dataset | Key findings | Gap addressed in this study |
|---|---|---|---|
| Buhalis & Sinarta (2019); Ivanov & Webster (2019); Melián-González et al. (2021) — Travel AI business value | Service-automation frameworks and consumer adoption modeling ($N$ = 1,476) in tourism and hospitality | Real-time personalization and low interaction effort drive chatbot booking conversion and ROI | Measures perceptions via surveys rather than instrumenting turn economy, slot redundancy, or policy compliance in LLM dialogues (RQ3, RQ4) |
| Jannach et al. (2021); Inaba et al. (2022) — Conversational recommenders & traveler styles | CRS architectural taxonomy; collection and analysis of multi-turn travel agency dialogues | Travelers differ in initiative, exploration breadth, and constraint rigidity; static policies cause repetitive turns | Describes traveler diversity qualitatively without unsupervised archetype clustering or prefix routing (RQ1, RQ2) |
| El Asri et al. (2017); Budzianowski et al. (2018); Zang et al. (2020) — Dialogue corpora | Microsoft Frames (1,369 package negotiations) and MultiWOZ 2.2 (10,438 multi-domain dialogues) | Frames captures nonlinear frame-switching and budget trade-offs; MultiWOZ standardizes slot tracking | Built for supervised state tracking; neither clusters user archetypes nor evaluates few-shot leakage or policy rules (RQ1–RQ4) |
| Reimers & Gurevych (2019); Lee et al. (2025); McInnes et al. (2017, 2018); Hennig (2007) — Embeddings & clustering | Sentence-BERT (384-d), Gemini embeddings (3,072-d), HDBSCAN, UMAP, and bootstrap ARI stability | Cosine geometry supports semantic clustering; bootstrap ARI and Silhouette quantify partition stability | Clusters static texts rather than prefix-truncated user trajectories paired with LLM behavioral ratings (RQ1) |
| Brown et al. (2020); Liu et al. (2022); Johnson et al. (2021) — In-context learning & FAISS | Few-shot prompting; $k$-NN exemplar selection (KATE); FAISS vector indexing | Nearest-neighbor in-context examples outperform random static examples on single-turn NLP tasks | Retrieves from one global pool without archetype partitioning, slot masking, or style/leakage evaluation (RQ2, RQ3) |

| Study and stream | Methodology and dataset | Key findings | Gap addressed in this study |
|---|---|---|---|
| Rebedea et al. (2023); Yao et al. (2024); Acikgoz et al. (2025); Gilardi et al. (2023) — Guardrails & evaluation | NeMo Guardrails; τ-bench policy benchmark; TD-EVAL metrics; LLM annotation reliability | Prompt-only LLMs violate domain rules; programmable rails and turn-level factual audits are essential | Does not test the 2 × 2 factorial interaction between declarative rules and cluster-adaptive style prompting (RQ3) |

*Note.* CRS = conversational recommender system; ARI = adjusted Rand index; FAISS = Facebook AI Similarity Search.

**Thematic Synthesis and Research Gaps**

*Business context: Hospitality service automation and conversational friction.* In digital travel and hospitality commerce, competitive differentiation depends on real-time itinerary co-creation ('nowness' service; Buhalis & Sinarta, 2019). Ivanov and Webster (2019) showed that automated service agents generate financial value only when they resolve consultative inquiries without human escalation, while Melián-González et al. (2021) demonstrated that consumer intention to use travel chatbots depends on interaction efficiency and task fit. Empirical studies of travel agency dialogues confirm that human consultants adapt their pacing—confirming bookings rapidly for decisive buyers while presenting side-by-side comparisons for exploratory shoppers (Inaba et al., 2022; Jannach et al., 2021).

*Task-oriented corpora and unsupervised trajectory clustering.* While MultiWOZ (Budzianowski et al., 2018) and MultiWOZ 2.2 (Zang et al., 2020) benchmark slot tracking across local services, they lack flight routing and multi-package price negotiation. Microsoft Frames (El Asri et al., 2017) directly captures multi-turn flight-and-hotel negotiation under budget constraints, making it the ideal primary corpus and MultiWOZ 2.2 an external structural comparison. Representing dialogues with Sentence-BERT (Reimers & Gurevych, 2019) and Gemini embeddings (Lee et al., 2025) enables rigorous comparison of K-Means, Ward agglomerative clustering, and HDBSCAN (McInnes et al., 2017) using Silhouette (Rousseeuw,

1987), Davies–Bouldin (Davies & Bouldin, 1979), bootstrap ARI stability (Hennig, 2007; Hubert & Arabie, 1985), and UMAP (McInnes et al., 2018; Pedregosa et al., 2011), augmented by reliable zero-temperature LLM behavioral ratings (Gilardi et al., 2023).

*Dynamic in-context learning, neuro-symbolic guardrails, and research gaps.* Few-shot prompting (Brown et al., 2020) and similarity-based FAISS retrieval (Johnson et al., 2021; Liu et al., 2022) improve single-turn task accuracy but risk entity contamination and policy drift in multi-turn dialogues. Programmable guardrails (Rebedea et al., 2023), policy benchmarks (τ-bench; Yao et al., 2024), and turn-level evaluation frameworks (TD-EVAL and BERTScore; Acikgoz et al., 2025; Zhang et al., 2020) motivate decoupling style from policy verification. Four research gaps follow: (1) discovering early-routable travel user archetypes free of outcome confounding (RQ1); (2) testing archetype-matched, slot-masked exemplar retrieval for style convergence (RQ2); (3) isolating the 2 × 2 factorial interaction between declarative rule verification and adaptive prompting (RQ3); and (4) quantifying multi-turn turn economy and redundant elicitation reductions across traveler personas (RQ4).

## Materials and Methods

### Data Sources, Access, and Ethical Governance

This study uses two public English dialogue corpora (Table 3): Microsoft Frames (El Asri et al., 2017; 1,369 raw dialogues, 1,334 analytic) and MultiWOZ 2.2 (Zang et al., 2020; seeded sample of 1,500 travel dialogues, 1,499 analytic). Both datasets were collected with research consent, use anonymized role-player IDs, and contain no personally identifiable information.

**Table 3**

*Dataset Overview*

| Attribute | Microsoft Frames (primary corpus) | MultiWOZ 2.2 (comparison corpus) |
|---|---|---|
| Records | 1,369 raw; 1,334 analytic (golden $D_{gold}$ = 933, held-out $D_{test}$ = 401) | 10,438 raw; travel sample of 1,500; 1,499 analytic |
| Domain setting | Multi-turn flight + hotel package negotiation under budget and date constraints | Multi-domain city services (hotel, train, taxi, attraction) without flights |
| Annotations used | Utterance text, speaker, dialogue acts, slot–value arguments, wizard task success, user rating | Utterance text, speaker, active services, turn-level slot state dictionaries |
| Analytical role | RQ1 clustering & refined archetypes, RQ2 prompting, RQ3 2×2 experiment, RQ4 2×3 simulation | External descriptive comparison in EDA and cross-corpus projection check in RQ1 |

## Data Dictionary and Data Cleaning

Both corpora were normalized into a two-tier Pydantic schema (*src/canonical_schema.py*): Tier 1 stores nine universal travel slots (origin, destination, departure/return dates, duration in nights, adults, children, budget in USD, and hotel tier), and Tier 2 stores five policy slots evaluated by the rule engine (visa window in days, passport validity in months, package type, room sharing, and discount percentage). Table 4 defines the analytical variables and units (raw source fields and canonical schema fields are listed in Appendix A, Table A3). Table 5 documents the programmatic quality audit and cleaning log. To prevent data leakage, the 1,334 analytic Frames dialogues were split once (seed 42, stratified on task success) into a 70% golden discovery partition ($D_{gold}$, $n$ = 933) used to fit clusters, PCA transformations, routers, and FAISS indices, and a 30% held-out test partition ($D_{test}$, $n$ = 401) used strictly for out-of-sample evaluation.

**Table 4**

*Data Dictionary of Analytical Variables With Units and Operational Definitions*

| Variable | Role | Type and unit | Operational definition and scale | RQ |
|---|---|---|---|---|
| Clustering algorithm | Primary IV | Categorical (3 levels) | K-Means ($k$=3), Ward agglomerative ($k$=3), HDBSCAN | 1 |
| Silhouette & Davies–Bouldin | Primary/Sec. DV | Continuous (unitless / ratio) | Cosine Silhouette [−1, 1] and cluster scatter ratio (≥ 0) | 1 |
| Adjusted Rand index (ARI) | Diagnostic DV | Continuous [−1, 1] (unitless) | Chance-corrected partition stability across 80% subsamples | 1 |
| LLM behavioral profile (16 fields) | Clustering / Audit | 10 ordinal (1–5) + 6 cat./bool. | 10 Likert scales + 3 goal/tone clustering fields + 3 audit fields (Table C1) | 1, 4 |
| Refined user archetype | IV / Moderator | Categorical (3 levels) | *Decisive Booker*, *Open Explorer*, *Rigid-Constraint Seeker* | 1–4 |
| Prompting strategy | Primary IV | Categorical (2 or 3 levels) | Zero-Shot, Static Few-Shot ($k$=3), Cluster-Adaptive Few-Shot ($k$=3) | 2–4 |
| Style convergence & Pacing err. | Primary/Sec. DV | Cosine [−1, 1] / Words (count) | Cosine to target archetype agent centroid; $\lvert words_{gen} - \mu_{words,a} \rvert$ | 2, 3 |
| Rule-engine mode | Primary IV | Binary (2 levels) | Rules Off (unguarded) vs. Rules Active (pre-check + verification) | 3 |
| Rule adherence rate (RAR) | Primary DV | Proportion [0, 1] (rate) | Share of violated policy rules enforced (and compliant states unblocked) | 3 |
| Exemplar leakage rate (ELR) | Primary DV | Proportion [0, 1] (rate) | Share of response entities originating from prompt exemplars | 3 |
| Factual slot coverage $F_1$ | Primary DV | Proportion [0, 1] ($F_1$ score) | Harmonic mean of precision and recall over active booking state slots | 3 |
| Turns to goal (log) & RER | Primary/Sec. DV | Log count (ln turns) / Rate [0, 1] | Log turns until [ACCEPT_BOOKING]; redundant slot requests / agent turns | 4 |

*Note.* IV = independent variable; DV = dependent variable; RER = redundant elicitation rate.

Raw and canonical fields are in Table A3.

**Table 5**

*Data Cleaning and Quality Audit Log*

| Issue | Variables | Detection | Frames | MultiWOZ | Treatment and rationale |
|---|---|---|---|---|---|
| Duplicate full text | Utterances | SHA-256 text hash | 0 | 1 | Removed duplicate to prevent double-weighting |
| Fewer than 3 turns | total_turns | Turn count < 3 | 35 | 0 | Excluded; minimum 3 turns needed for prefix routing |
| Turn-count outliers | total_turns | > Q3 + 1.5 × IQR | 22 (> 32.5) | 13 (> 27.0) | Retained as genuine negotiations; log-transformed in RQ4 |

| Issue | Variables | Detection | Frames | MultiWOZ | Treatment and rationale |
|---|---|---|---|---|---|
| Unobserved slots | Tier 1 & 2 slots | Never mentioned | Table E1 | Table E1 | Preserved as null (None); never imputed |
| Entity values in text | Utterance text | Slot/number regex | All | All | Masked with [SLOT] tokens in exemplars to prevent leakage |
| Analytic sample | All fields | Post-cleaning count | 1,334 (933/401) | 1,499 | Locked with stratified 70/30 split (seed 42) |

*Note.* Zero duplicate IDs or empty utterances occurred in either corpus. Missing user ratings (11 in Frames) were retained.

## Exploratory Data Analysis (Univariate, Bivariate, and Multivariate)

### *Univariate Analysis*

Table 6 and Figure 1 summarize the univariate distributions. Frames and MultiWOZ 2.2 have similar mean lengths ($M$ = 14.93 vs. 15.17 turns), but Frames exhibits greater dispersion ($SD$ = 6.74 vs. 4.46) and a heavier right tail due to multi-offer package comparisons. Golden and held-out partitions are nearly identical across all metrics. Slot completeness (Appendix E, Table E1 and Figure E1) confirms high coverage of origin (98.4%), destination (99.1%), dates (96.0%), and budget (78.5%) in Frames, but 0.0% coverage of visa, passport, and discount rules in both corpora, motivating synthetic Tier-2 injection in RQ3.

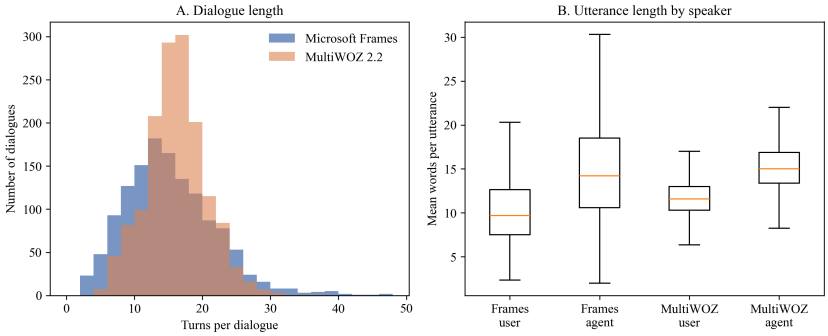
**Table 6**

*Univariate Descriptive Statistics of the Analytic Samples*

| Sample | *N* | Turns *M* (*SD*) | *Mdn* | Range | User words | Agent words | Task success (%) |
|---|---|---|---|---|---|---|---|
| Microsoft Frames | 1,334 | 14.93 (6.74) | 14 | 3–48 | 10.6 | 15.0 | 94.2 |
| MultiWOZ 2.2 | 1,499 | 15.17 (4.46) | 16 | 4–36 | 11.7 | 15.1 | — |
| Frames golden partition (D_gold) | 933 | 14.95 (6.75) | 14 | 3–48 | 10.7 | 15.3 | 94.2 |
| Frames held-out partition (D_test) | 401 | 14.90 (6.71) | 14 | 3–43 | 10.4 | 14.5 | 94.3 |

**Figure 1**

*Univariate Distribution of Dialogue Length by Corpus*




*Note.* Frames $N = 1{,}334$; MultiWOZ $N = 1{,}499$.

### *Bivariate Analysis*

Parametric (Pearson $r$) and non-parametric (Spearman ρ) bivariate correlations across all 1,334 Frames dialogues (Table 7; Figure 2) show that dialogue length is strongly driven by slot revisions ($r = .74$, $\rho = .76$, $p < .001$) and agent package offers ($r = .56$, $\rho = .61$, $p < .001$). Linking structural counts to LLM behavioral ratings confirms construct validity: slot revisions correlate strongly with user exploration breadth ($r = .60$, $\rho = .67$, $p < .001$), whereas decisiveness correlates negatively with dialogue length ($r = -.31$, $\rho = -.29$, $p < .001$).

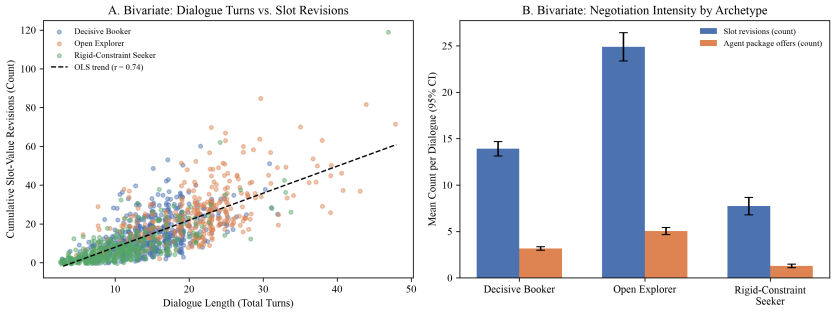
**Table 7**

*Bivariate Parametric and Non-Parametric Correlations of Dialogue Dynamics (Frames)*

| Variable pair | Pearson $r$ | $p$ (Pearson) | Spearman ρ | $p$ (Spearman) |
|---|---|---|---|---|
| Total Turns vs. Slot Revisions | .741 | < .001 | .759 | < .001 |
| Total Turns vs. Agent Package Offers | .561 | < .001 | .608 | < .001 |
| Total Turns vs. User Hesitation Markers | .336 | < .001 | .355 | < .001 |
| Slot Revisions vs. Exploration Breadth Rating | .603 | < .001 | .673 | < .001 |
| Total Turns vs. User Decisiveness Rating | −.310 | < .001 | −.289 | < .001 |
| User Words/Turn vs. Information Up Front Rating | .158 | < .001 | .206 | < .001 |
| Agent Offers vs. Destination Flexibility Rating | .324 | < .001 | .358 | < .001 |

**Figure 2**

*Bivariate Relationships Between Dialogue Length, Slot Revisions, and Archetype Negotiation Intensity*



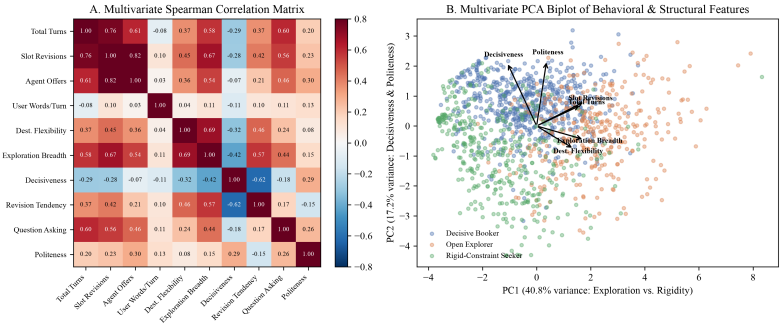

*Note.* Panel A: Turns vs. slot revisions with OLS trend. Panel B: Mean revisions and package offers by refined archetype (95% CI).

### *Multivariate Analysis*

Multivariate structure across ten structural and behavioral variables was examined via Spearman rank correlation and principal component analysis (PCA; Figure 3; full loadings in Appendix E, Table E2). The first three components explain 68.6% of total variance: PC1 (40.8%) captures *Exploration vs. Rigidity* (revisions .42, turns .39, exploration breadth .39), PC2 (17.2%) captures *Decisiveness & Politeness* (−.51, −.46), and PC3 (10.6%) isolates *User Verbosity* (.89), separating the three refined archetypes across a continuous behavioral manifold.

**Figure 3**

*Multivariate Spearman Correlation Matrix and Principal Component Biplot of Behavioral Features*




*Note.* Panel A: Rank correlation matrix. Panel B: PCA biplot colored by refined archetype with top loading vectors.

## Research Design, Hypotheses, and Statistical Power Analysis

Sample sizes were computed a priori for each planned analysis using Cohen's medium effect $f = .25$ (Cohen, 1988), $\alpha = .05$, and target power $1 - \beta \geq .80$ via the noncentral $F$ distribution (Faul et al., 2007):

$$1 - \beta = 1 - F_{nc}(F_{crit};\, df_1,\, df_2,\, \lambda), \quad F_{crit} = F_{1-\alpha}(df_1,\, df_2) \qquad (1)$$

where $\lambda = f^2 N$ for between-subjects designs (RQ1, RQ3, RQ4) and $\lambda = f^2 N m \varepsilon / (1 - \rho)$ for repeated-measures RQ2 ($m = 3$, $\rho = .50$, $\varepsilon = 1$). Minimum sample sizes were rounded to balanced cells with a 10% buffer (Table 8), yielding final power of .805–.878. In RQ1, because each of the 53 random seeds draws $n = 500$ dialogues without replacement from $D_{gold}$ ($n = 933$, a 53.6% finite-population fraction), we pair the 159-run one-way benchmark with a matched-subsample repeated-measures ANOVA and Friedman test across the 53 shared seeds plus full-$D_{gold}$ point estimates. In RQ3, 36 stratified held-out dialogues (18 policy-violating, 18 compliant; 12 per archetype) are evaluated across all four 2 × 2 cells ($N = 144$ evaluations). In RQ4, **180 distinct, non-overlapping held-out seed dialogues** (60 per archetype, partitioned into $n = 30$ unique

dialogues for *Zero-Shot* and $n = 30$ unique dialogues for *Cluster-Adaptive Few-Shot*) are allocated across the six 2 × 3 cells, ensuring strict between-subjects independence ($df_{error} = 174$).

**Table 8**

*Sample Size and Power Calculations Matched to Each Planned Analysis*

| RQ | Planned analysis | *df* at $N_{min}$ | λ | $F_{crit}$ | $N_{min}$ | Power | $N_{final}$ (per group) | Final power |
|---|---|---|---|---|---|---|---|---|
| RQ1 | One-way ANOVA (3 algorithms) | 2, 155 | 9.88 | 3.05 | 158 | .802 | 159 (53) | .805 |
| RQ2 | One-way repeated-measures ANOVA (3 conditions) | 2, 54 | 10.50 | 3.17 | 28 | .812 | 33 (11) | .878 |
| RQ3 | 2 × 2 factorial ANOVA | 1, 124 | 8.00 | 3.92 | 128 | .801 | 144 (36) | .846 |
| RQ4 | 2 × 3 factorial ANOVA | 2, 152 | 9.88 | 3.06 | 158 | .802 | 180 (30) | .855 |

## Model Selection, Mathematical Formulation, and Evaluation Metrics

*Trajectory and hybrid user representations (RQ1).* Primary trajectory vectors pool de-lexicalized turn embeddings $\mathbf{e}_t$ with exponential decay ($\lambda_d = 0.15$):

$$\mathbf{v} = \Sigma_{t=1}^{T} w_t \mathbf{e}_t, \quad w_t = \exp[-\lambda_d(T - t)] \,/\, \Sigma_{j=1}^{T} \exp[-\lambda_d(T - j)] \quad (2)$$

To prevent end-of-dialogue outcome confounding, the refined archetype representation (*Blocks*) concatenates two L2-normalized blocks computed strictly on the first $M$ user messages: (1) the top 20 principal components (fitted on $D_{gold}$) of the 3,072-dimensional Gemini embedding (*gemini-embedding-001*; Lee et al., 2025), and (2) a standardized 16-field LLM behavioral schema comprising 13 clustering fields (ten 1–5 ordinal rating scales plus primary goal, secondary goals, and emotional tone, encoded into 26 numeric columns) and three audit-only fields (session outcome, party type, and unusual-conversation flag, excluded from clustering input; Table C1; test–retest weighted κ = .87–.96 across the 10 ordinal scales and κ = .75–.98 across the 6 categorical/boolean fields; revision correlation ρ = .73, Table C2). In RQ2–RQ4, *RefinedArchetypeRouter* projects each test dialogue's opening user prefix into this golden-fitted 46-dimensional *Blocks* space, predicts the user archetype via L1-regularized logistic regression,

and applies a combined adaptation package (retrieving $k = 3$ slot-masked exemplars from that archetype's FAISS sub-index alongside an explicit archetype pacing/action directive). In RQ3, the deterministic rule engine (*src/rule_engine.py*, 14/14 boundary tests passed) evaluates five YAML policy rules (Table 9) within an integrated governance pipeline combining pre-generation rule injection, [SLOT] exemplar masking, and post-generation regex sanitization. In RQ4, each closed-loop simulation pairs the *Verified Informed Slot Ledger* with archetype-specific turn-economy policies, counting total conversational turns (user plus assistant messages) until the simulated traveler issues <code>[ACCEPT_BOOKING]</code> (subject to a protocol minimum of 5 total turns: <code>U1 → A1 → U2 → A2 → U3</code> once budget is confirmed). Table 10 defines all evaluation metrics.

**Table 9**

*Declarative Policy Rules Used in RQ3*

| Rule ID and name | Trigger condition | Action | Severity |
|---|---|---|---|
| RULE_VISA_WINDOW: Visa Processing Lead Time | departure_lead_days < 15; is_international = true | block and flag | fatal |
| RULE_PASSPORT_VALIDITY: Passport Validity Threshold | passport_expiry_months < 6; is_international = true | block and flag | fatal |
| RULE_PAX_CAPACITY: Maximum Adult Capacity per Single Booking | traveler_count_adults > 8; package_type = 'FIT' | convert to group | warning |
| RULE_DISCOUNT_THRESHOLD: Maximum Discretionary Sales Discount | discount_applied_percentage > 15.0 | clamp discount | fatal |
| RULE_BUDGET_MINIMUM: Minimum Feasible Per-Pax International Budget | budget stated; budget_amount < 500; is_international = true | prompt realistic budget | warning |

**Table 10**

*Evaluation Metrics, Mathematical Formulae, and Interpretation*

| Metric | Formula | RQ | Interpretation |
|---|---|---|---|
| Silhouette & DB index | $s(i) = [b(i)-a(i)]/\max\{a,b\}$; $DB = (1/k)\Sigma \max(\sigma_i+\sigma_j)/d(\mathbf{c}_i,\mathbf{c}_j)$ | 1 | Higher Silhouette [−1, 1] and lower DB (≥ 0) indicate better separation |
| ARI & Cohen's κ | $ARI = (RI-\mathrm{E}[RI])/(\max RI-\mathrm{E}[RI])$; $\kappa = (p_o-p_e)/(1-p_e)$ | 1, 3, 4 | Bootstrap 80%-subsample stability (ARI ≥ .60) and routing/judge agreement |
| Style cosine & Pacing err. | $\cos(\mathbf{r}, \mathbf{c}_a) = (\mathbf{r}\cdot\mathbf{c}_a)/(\|\mathbf{r}\|\|\mathbf{c}_a\|)$; $\lvert W_{\text{gen}} - \mu_{W,a}\rvert$ | 2, 3 | De-lexicalized SBERT structural style cosine and word-count pacing error |
| Rule adherence (RAR) | Sens. = $\lvert R_{\text{enf}} \cap R_{\text{viol}}\rvert/\lvert R_{\text{viol}}\rvert$ ($n$=18/cell); Spec. = 1 − false_block ($n$=18/cell); Overall ($n$=36/cell) | 3 | Separated into Violation Enforcement (Sensitivity), Compliant Acceptance (Specificity), and Overall Accuracy |
| Exemplar leakage (ELR) | $\lvert E_{\text{resp}} \cap (E_{\text{ex}} \setminus E_{\text{diag}})\rvert$ / $\max(1, \lvert E_{\text{resp}}\rvert)$ | 3 | Proportion [0, 1] of response entities leaked from prompt exemplars |
| Slot coverage $F_1$ | 2 · Precision · Recall / (Precision + Recall) over active state slots | 3 | Proportion [0, 1] of active booking slots grounded in response |
| Turns to goal & RER | TGC = total turns (user + assistant messages) to [ACCEPT_BOOKING] (min = 5); RER = $A_{\text{redundant}}$ / $A_{\text{total}}$ | 4 | Lower total/log conversational turns and lower RER indicate superior turn economy |

*Note.* Parametric ANOVAs are paired with Welch/HC3 and non-parametric (Kruskal–Wallis, Friedman, Mann–Whitney, ART; Wobbrock et al., 2011) checks.

## Architecture and Workflow

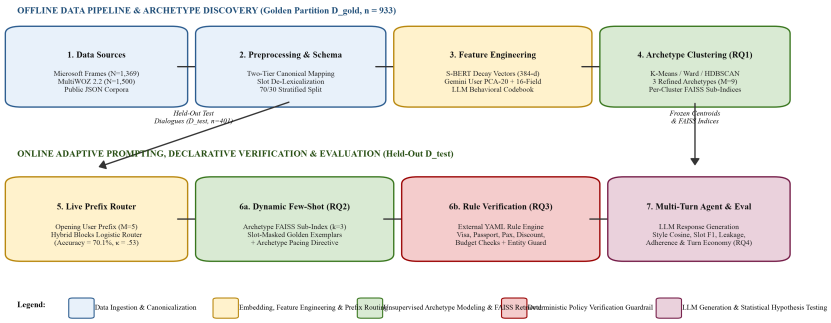
### System Architecture and Algorithmic Workflows

Figure 4 illustrates the five-layer neuro-symbolic system architecture, and Figure 5 details the step-by-step execution workflows for RQ1 through RQ4. Offline discovery and indexing (Layers 1–2) operate strictly on the golden partition ($D_{\text{gold}}$, $n$ = 933): raw dialogues are parsed into two-tier Pydantic trajectories, prefix embeddings (*gemini-embedding-001*) and 16-field LLM behavioral profiles are combined into the *Blocks* representation on the first $M$ = 9 user messages, and three $K$-Means centroids, three archetype-specific FAISS *IndexFlatIP* sub-indices,

and the L1-regularized prefix router are frozen. Online runtime and evaluation (Layers 3–5) operate on held-out test dialogues ($D_{\text{test}}$, $n = 401$): incoming prefixes are projected into the golden *Blocks* space and routed to their predicted archetype, $k = 3$ slot-masked exemplars are retrieved, the Verified Informed Slot Ledger and declarative YAML rule engine evaluate state constraints, *gemini-2.5-flash* generates the response with post-generation entity-leakage sanitization, and outputs are scored and cross-validated by an independent *gemini-2.5-pro* judge.

**Figure 4**

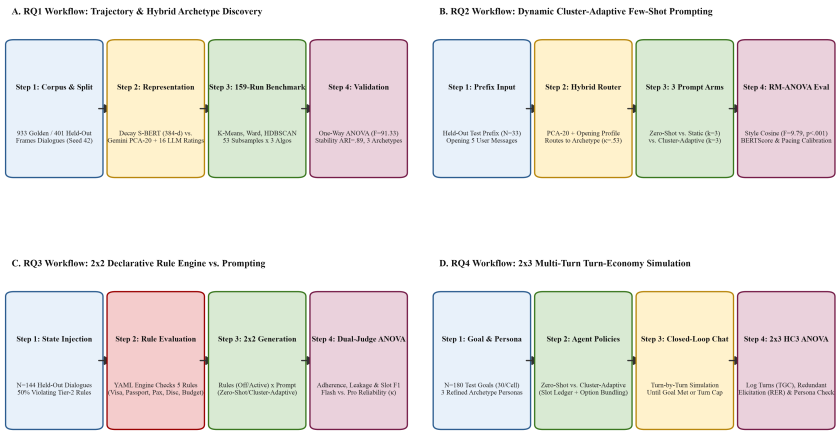
*End-to-End Neuro-Symbolic Cluster-Adaptive Conversational System Architecture*




*Note.* Offline fitting uses $D_{\text{gold}}$ ($n = 933$); online routing, retrieval, rule verification, and evaluation use $D_{\text{test}}$ ($n = 401$).

**Figure 5**

*Algorithmic Execution Workflows Across Research Questions RQ1 to RQ4*




*Note.* Step-by-step experimental pipelines and statistical hypothesis tests for RQ1 (Panel A), RQ2 (Panel B), RQ3 (Panel C), and RQ4 (Panel D).

### Pipeline Component Specifications and Technology Stack

The software stack is implemented in Python 3.11 across *src/interim_pipeline/* and *src/final_pipeline/* using *pandas* 2.2, *numpy* 1.26, and *pydantic* 2.9 for data validation; *sentence-transformers* (*all-MiniLM-L6-v2*), Google *genai* SDK (*gemini-embedding-001*, *gemini-2.5-flash*, *gemini-2.5-pro*), and *faiss-cpu* for vector indexing and generation; *scikit-learn* 1.5, *hdbscan*, and *umap-learn* for clustering; *scipy.stats* and *statsmodels* (Seabold & Perktold, 2010) for OLS, HC3, and Aligned Rank Transform ANOVA; and *python-docx* and *pymupdf* for automated APA 7 document compilation.

## Results, Analysis, and Synthesis

### Consolidated Model Performance and Hypothesis Testing Summary

Table 11 summarizes the empirical results, parametric test statistics, heteroscedasticity-robust (Welch/HC3) and non-parametric (Kruskal–Wallis, Friedman, Mann–Whitney, and

Aligned Rank Transform [ART]) checks, effect sizes, and hypothesis decisions across RQ1–RQ4. All four null hypotheses specified for the study ($H_{01}$–$H_{04}$) are rejected at $p < .001$ on the refined user archetype pipeline.

**Table 11**

*Consolidated Summary of Empirical Results and Hypothesis Outcomes Across RQ1–RQ4*

| RQ and outcome | Sample & design | Baseline vs. proposed performance | Primary parametric test & effect size | Robust / non-parametric check | Outcome |
|---|---|---|---|---|---|
| RQ1: Algorithm separation & refined archetypes | $N = 159$ runs (53 matched $n = 500$ draws from $D_{gold} = 933$; $D_{test} = 401$) | MiniLM Sil: Perm = −.013 vs. KM = .147, Ward = .118, HDB = .211 (39.1% noise); Blocks ARI = .89 (.87 abl.); routing κ = .535 ($m$=5) to .958 ($m$=9) | One-way $F(2, 156) = 91.33$, $p < .001$, $\eta^2 = .54$; Matched RM-ANOVA $F(1.72, 89.30) = 83.04$, $p < .001$, $\eta_p^2 = .615$ | Friedman $\chi^2(2) = 56.30$, Welch $F = 66.98$, Kruskal $H(2) = 86.82$, all $p < .001$ | Reject $H_{01}$ |
| RQ2: De-lexicalized SBERT style cosine (Refined) | $N = 33$ dialogues × 3 conditions (99 calls; combined exemplars + directive) | Static = .453, Zero-Shot = .530 vs. Cluster-Adaptive = .597 (Gemini/BERTScore null, $p \geq .315$) | RM-ANOVA $F(1.88, 60.29) = 9.79$, $p < .001$, $\eta_p^2 = .23$ | Friedman $\chi^2(2) = 12.50$, $p = .002$; Holm $p < .001$ vs. Static | Reject $H_{02}$ |
| RQ2: Pacing calibration error (words) | $N = 33$ dialogues × 3 conditions (99 calls) | Zero-Shot = 10.12w, Static = 6.33w vs. Cluster-Adaptive = 3.87w | RM-ANOVA $F(1.30, 41.70) = 7.10$, $p = .007$, $\eta_p^2 = .18$ | Friedman $\chi^2(2) = 14.27$, $p < .001$; Holm $p = .004$ vs. Zero-Shot | Reject $H_{02}$ |
| RQ3: Policy rule adherence (Sens. / Spec. / Overall) | $N = 144$ (2 × 2 across 36 dialogues, $n = 36$/cell; $\kappa_{judge} = .917$) | Viol. Sens. ($n$=18/cell): 5.6% → 98.6%; Comp. Spec. ($n$=18/cell): 94.4% vs. 88.9%; Overall ($n$=36/cell): .500 → .938 | Sens. $F(1, 68) = 498.78$, $p < .001$, $\eta_p^2 = .880$; Overall Rules $F(1, 140) = 43.92$, $p < .001$, $\eta_p^2 = .24$ | HC3 $F = 42.71$, $p < .001$; ART $F = 39.12$, $p < .001$ | Reject $H_{03}$ |
| RQ3: Exemplar entity leakage & Slot $F_1$ | $N = 144$ (2 × 2, $n = 36$/cell) | Leakage: Rules Off Cluster = .359 (55.6% cases) vs. Rules Active Cluster = .000 (0.0%); Slot $F_1$: .485 → .952 | Leakage Int. $F(1, 140) = 34.82$, $p < .001$; Slot $F_1$ Rules $F = 489.52$, $p < .001$ | Leakage Sum-HC3 $F = 33.85$, ART Int. $F = 2,148.81$, Simple Welch $t(35) = 5.90$, all $p < .001$ | Reject $H_{03}$ |

| RQ and outcome | Sample & design | Baseline vs. proposed performance | Primary parametric test & effect size | Robust / non-parametric check | Outcome |
|---|---|---|---|---|---|
| RQ4: Log turns to goal completion & RER | $N = 180$ unique dialogues (2 × 3, $n$ = 30/cell; $\kappa_{persona}$ = .933) | Total turns reduced −34.2% (DB), −29.5% (OE), −42.7% (RCS; 5.00-turn floor); RER .109 → .000 | Prompt $F(1, 174) = 664.43$, $p < .001$; Int. $F(2, 174) = 10.89$, $p < .001$, $\eta_p^2 = .11$ | HC3 Int. $F = 8.58$, $p < .001$; ART Int. $F = 209.46$, $p < .001$ | Reject $H_{04}$ |

*Note.* DB = Decisive Booker; OE = Open Explorer; RCS = Rigid-Constraint Seeker; Perm = permutation baseline; Sens. = violation enforcement sensitivity; Spec. = compliant acceptance specificity.

**RQ1 Results: Trajectory Clustering, Refined Archetypes, and Prefix Routing**

*Primary algorithm benchmark, resampling sensitivity, and embedding invariance.* Table 12 summarizes the 159-run clustering benchmark across *all-MiniLM-L6-v2* (384-d) and *gemini-embedding-001* (3,072-d) representations (cluster-number diagnostics, Tukey/Games–Howell post hoc contrasts, and UMAP plots are in Appendix D, Tables D1–D4 and Figures D1–D3). Because each of the 53 random seeds draws $n = 500$ dialogues without replacement from the finite golden partition ($D_{gold}$, $n = 933$, a 53.6% sampling fraction with ~268 overlapping dialogues between draws), the 159 evaluations represent a finite-population resampling stability analysis rather than 159 independent samples; accordingly, we report both matched-subsample repeated-measures tests across the 53 shared seeds and full-$D_{gold}$ point estimates alongside the standard one-way benchmark. On MiniLM embeddings, HDBSCAN achieved the highest Silhouette on its retained non-noise subset ($M = .211$, $SD = .044$), followed by $K$-Means ($M = .147$, $SD = .019$; full-$D_{gold}$ point estimate = .141, Davies–Bouldin = 3.244) and Ward agglomerative clustering ($M = .118$, $SD = .040$), all exceeding the label-shuffled permutation baseline ($M = -.013$). Algorithm differences were highly significant under the matched-subsample repeated-measures ANOVA, $F(1.72, 89.30) = 83.04$, $p < .001$, $\eta_p^2 = .615$, the non-

parametric paired Friedman test, $\chi^2(2) = 56.30$, $p < .001$, the one-way ANOVA, $F(2, 156) = 91.33$, $p < .001$, $\eta^2 = .54$, Welch's test, $F(2, 89.15) = 66.98$, $p < .001$, and Kruskal–Wallis $H(2) = 86.82$, $p < .001$. Replicating on 3,072-d Gemini embeddings reproduced the same ordering with stronger separation: for turn-pooled vectors (*G-pooled*), *K*-Means = .158, Ward = .150, and HDBSCAN = .289 ($F(2, 156) = 77.09$, $\eta^2 = .50$, $p < .001$); for document vectors (*G-doc*), *K*-Means = .161, Ward = .213, and HDBSCAN = .345 ($F(2, 156) = 127.47$, $\eta^2 = .62$, $p < .001$). Critically, however, HDBSCAN's higher Silhouette is computed strictly after excluding unassigned noise points: HDBSCAN failed at default settings (*min_cluster_size* = 15) in 53/53 runs and discarded **39.1%** (*SD* = 12.7%) of MiniLM dialogues, **35.8%** of Gemini turn-pooled dialogues, and **25.7%** of Gemini document dialogues as noise when relaxed. Because a live travel assistant cannot refuse to route 26%–39% of incoming customer sessions, *K*-Means ($k = 3$) was selected on operational grounds to guarantee **100% dialogue coverage** and deterministic Voronoi centroid routing. Golden centroids generalized to held-out Frames dialogues ($t(754.9) = 0.42$, $p = .676$) but differed from MultiWOZ 2.2 ($t(576.2) = 38.29$, $p < .001$; Table D4).

**Table 12**

*Cluster Quality by Algorithm and Embedding Family Across 53 Matched Subsample Seeds (159 Runs)*

| Representation and algorithm | *n* seeds | Retained (%) | Silhouette *M* (*SD*) | Davies–Bouldin *M* (*SD*) | Omnibus & matched RM tests ($p < .001$) |
|---|---|---|---|---|---|
| MiniLM-L6-v2 (384-d) — K-Means ($k = 3$) | 53 | 100.0% | .147 (.019) | 3.129 (0.113) | One-way $F = 91.33$, $\eta^2 = .54$ |
| MiniLM-L6-v2 (384-d) —Ward Agglomerative | 53 | 100.0% | .118 (.040) | 3.550 (0.269) | Matched RM $F = 83.04$, $\eta_p^2 = .615$ |
| MiniLM-L6-v2 (384-d) — HDBSCAN (non-noise) | 53 | 60.9% | .211 (.044) | 2.657 (0.241) | Friedman $\chi^2 = 56.30$, Welch $F = 66.98$ |

| Representation and algorithm | *n* seeds | Retained (%) | Silhouette *M* (*SD*) | Davies–Bouldin *M* (*SD*) | Omnibus & matched RM tests ($p < .001$) |
|---|---|---|---|---|---|
| MiniLM-L6-v2 (384-d) — Permutation baseline | 53 | 100.0% | −.013 (.006) | — | Full $D_{gold}$ KM Sil = .141, DB = 3.244 |
| Gemini-001 (3,072-d turn pool) — KM / Ward / HDB | 53 × 3 | 100 / 100 / 64.2% | .158 / .150 / .289 | 2.74 / 2.96 / 2.89 | $F = 77.09$, $\eta^2 = .50$; Kruskal $H = 67.31$ |
| Gemini-001 (3,072-d document) — KM / Ward / HDB | 53 × 3 | 100 / 100 / 74.3% | .161 / .213 / .345 | 3.14 / 3.09 / 2.42 | $F = 127.47$, $\eta^2 = .62$; Kruskal $H = 98.10$ |

*Note.* Each seed draws $n = 500$ dialogues without replacement from $D_{gold}$ ($n = 933$; 53.6% finite-population fraction). HDBSCAN metrics apply only to retained non-noise dialogues (discarding 39.1% noise on MiniLM, 35.8% on Gemini turn pool, and 25.7% on Gemini document vectors).

*Sensitivity analysis, choice of k = 3 on Blocks, refined user archetypes, and hybrid prefix routing.* Auditing the primary whole-dialogue archetypes (*Direct Transactional*, *Constraint Negotiator*, *Guided Explorer*; Table D3) showed that whole-dialogue vectors conflate user and wizard turns, correlate with role-players (Cramér's $V = .19$), and yield low early-turn routing agreement ($\kappa = .11$). Evaluating the 46-dimensional hybrid *Blocks* representation (PCA-20 Gemini user prefix embedding + 26 encoded numeric columns from the 13 clustering fields of the 16-field LLM behavioral schema) across truncation horizons $M \in \{3, 5, 7, 9, \text{whole}\}$ and cluster counts $k \in \{2, ..., 8\}$ (Appendix D, Tables D1, D5, and D6; Figure D4) demonstrated that truncating to the first $M = 9$ user messages at $\boldsymbol{k = 3}$ is uniquely optimal. Specifically, on the final 46-d *Blocks* ($M = 9$) representation (Table D1, Panel B), $k = 3$ achieves the **lowest (best) Davies–Bouldin index (2.647 vs. 2.796 at $k = 2$ and 2.729–2.784 at $k = 4$–6)**, highest eligible Silhouette (.094; null-calibrated .033 vs. .002–.010 at $k = 4$–6), high bootstrap resampling stability across 50 pairs of 80% golden subsamples (**ARI = .89** in the truncation grid [Table D5]; **ARI = .87** with 89.2% core membership at ≥ 90% co-assignment under the independent random seed of the 6-configuration ablation [Table D6]), and balanced cluster sizes (smallest cluster =

25.5% ≥ 20% floor). In contrast, $k = 2$ (DB = 2.796) collapses *Decisive Bookers* and *Rigid-Constraint Seekers* into a single undifferentiated bucket, while $k = 4$, 5, and 6 suffer a sharp drop in bootstrap stability (ARI = .589, .593, and .631) and fragment into small clusters (11.3%–12.5%, violating the 20% balanced-segment threshold). Since the Microsoft Frames dataset does not provide predefined behavioral archetype labels, we generated reference archetypes ($y^{(9)}$) through unsupervised *K*-Means ($k = 3$) clustering on the 46-dimensional hybrid *Blocks* representation ($X^{(9)}$) of the first nine user messages (see Appendix F and Table F1 for the deterministic centroid-naming protocol, and Appendix G and Table G1 for the distinct validation role of Judge LLMs). Importantly, total dialogue turns (*T*, user plus wizard turns, $M = 14.93$, $Mdn = 14.0$; 96.4% ≥ 5 turns and 83.2% ≥ 9 turns) must be distinguished from user-only messages (*M*, customer turns only, $M = 7.78$, $Mdn = 7.0$; 83.2% ≥ 5 user messages and 37.6% ≥ 9 user messages): offline clustering caps input at min(user_turns, 9) to strip late checkout/abandonment turns from long sessions while retaining all user messages for shorter sessions. We subsequently trained prefix-specific multinomial Logistic Regression classifiers on $D_{\text{gold}}$ ($n = 933$) to predict these nine-message reference cluster assignments using only the first $m \in \{3, 5, 7, 9\}$ user messages ($X^{(m)}$). Because raw percentage agreement ($p_o$) includes expected chance agreement ($p_e \approx 35.4\%–36.6\%$ given uneven cluster shares) and compares against an unsupervised cluster target rather than human labels, we report both $p_o$ and chance-corrected Cohen's $\kappa = (p_o - p_e) / (1 - p_e)$. On the held-out test partition ($D_{\text{test}}$, $n = 401$), chance-corrected prefix-to-cluster agreement increased from $\kappa = .359$ ($p_o = 59.4\%$, 238/401, $p_e = 36.6\%$; 87.5% on < 3 user messages) at $m = 3$ messages (≈ 5 total turns) to $\kappa = .535$ ($p_o = 70.1\%$, 281/401, $p_e = 35.7\%$; 81.8% on < 5 and 67.8% on ≥ 5 user messages) at $m = 5$ messages (≈ 9 total turns, the runtime cut-off used in RQ2) and $\kappa = .722$ ($p_o = 82.0\%$, 329/401, $p_e = 35.5\%$; 93.2% on < 7 and

74.6% on ≥ 7 user messages) at $m = 7$ messages (≈ 13 total turns). At $m = 9$ messages (≈ 17 total turns), agreement reached $\kappa = .958$ ($p_o = 97.3\%$, 390/401, $p_e = 35.4\%$; 96.0% on ≤ 9 and 99.3% on ≥ 9 user messages), representing a same-feature boundary-consistency check between the linear Logistic Regression hyperplane and the $K$-Means Voronoi centroids on $X^{(9)}$. These results demonstrate progressively stronger early-prefix agreement with the clustering-derived reference archetypes; they do not, by themselves, establish independent human validation of the archetypes. Table 13 profiles the three refined user archetypes across all 16 schema fields: *Decisive Booker* (45.0%; high decisiveness = 3.97, politeness = 4.26, 99.3% booking goal), *Open Explorer* (25.5%; high destination flexibility = 3.49 and exploration breadth = 2.96, longest dialogues = 20.78 turns), and *Rigid-Constraint Seeker* (29.5%; fixed destination/date constraints = 2.20/2.18, concise 10.60-turn sessions where 68.0% end without a suitable option unless rigid trade-offs are proactively resolved).

**Table 13**

*Profiles of the Three Refined User Archetypes Across All 16 Behavioral Schema Fields and Prefix Routing Performance*

| Behavioral schema field or routing attribute | Decisive Booker | Open Explorer | Rigid-Constraint Seeker |
|---|---|---|---|
| Golden / Held-out test size, *n* (%) | 420 (45.0%) / 175 | 238 (25.5%) / 99 | 275 (29.5%) / 127 |
| Dialogue turns / User messages, *M* | 14.49 (4.54) / 7.58 | 20.78 (6.95) / 10.73 | 10.60 (5.73) / 5.55 |
| 1. Destination flexibility (1–5 ordinal) | 2.54 | 3.49 | 2.20 |
| 2. Date flexibility (1–5 ordinal) | 2.57 | 2.77 | 2.18 |
| 3. Exploration breadth (1–5 ordinal) | 1.97 | 2.96 | 1.59 |
| 4. Decisiveness (1–5 ordinal) | 3.97 | 2.58 | 3.23 |
| 5. Detail orientation (1–5 ordinal) | 2.39 | 2.87 | 1.46 |
| 6. Revision tendency (1–5 ordinal) | 2.00 | 3.03 | 2.40 |
| 7. Question asking (1–5 ordinal) | 2.05 | 2.72 | 1.43 |
| 8. Politeness (1–5 ordinal) | 4.26 | 3.80 | 3.07 |
| 9. Price sensitivity (1–5 ordinal) | 3.20 | 3.18 | 3.12 |
| 10. Information given up front (1–5 ordinal) | 2.18 | 1.82 | 2.16 |

| Behavioral schema field or routing attribute | Decisive Booker | Open Explorer | Rigid-Constraint Seeker |
|---|---|---|---|
| 11. Primary goal: explore / book trip (%) | 0.5% / 99.3% | 39.1% / 50.8% | 2.2% / 89.8% |
| 12. Secondary goals present (%) | 42.4% | 56.3% | 16.4% |
| 13. Emotional tone: positive / neutral (%) | 52.4% / 46.7% | 21.0% / 69.3% | 7.6% / 59.6% |
| 14. Session outcome (audit): booked / no option (%) | 90.5% / 6.2% | 68.1% / 18.5% | 28.0% / 68.0% |
| 15. Party type (audit): solo or couple (%) | 37.1% | 44.5% | 34.5% |
| 16. Unusual conversation flag (audit, %) | 1.0% | 1.7% | 6.9% |
| Frames wizard task success (%) | 94.5% | 96.2% | 92.0% |

*Note.* Golden $N = 933$; held-out $N = 401$ ($\chi^2(2) = 0.93$, $p = .628$). Fields 1–13 are used in clustering (*llm_matrix*: 10 ordinal + 6 primary_goal + 6 secondary_goals + 4 emotional_tone = 26 numeric columns); Fields 14–16 are audit-only (excluded from clustering to prevent outcome leakage). Held-out *Blocks* prefix-to-cluster raw agreement $p_o$ (chance-corrected κ, $p_e \approx 35.4\%$–36.6%) against the $M$=9 $K$-Means reference assignment: $m$=3: 59.4% (.359); $m$=5: 70.1% (.535); $m$=7: 82.0% (.722); $m$=9 (same-feature boundary consistency): 97.3% (.958).

**RQ2 Results: Cluster-Adaptive Few-Shot Prompting on Primary Versus Refined Archetypes**

Table 14 compares the repeated-measures prompting experiment ($N = 33$ held-out dialogues × 3 conditions = 99 Gemini 2.5 Flash generations) on the coarse primary archetypes versus the refined user archetypes, and Figure 6 displays the refined distributions (Holm contrasts across all six outcomes are in Appendix E, Table E3). Importantly, the *Cluster-Adaptive Few-Shot* condition evaluates a **combined personalization package** that couples archetype-partitioned FAISS exemplar retrieval ($k = 3$ slot-masked exemplars from the routed sub-index) with an explicit archetype pacing directive in the system prompt. On coarse primary archetypes with an unprojected 3-turn router (39.4% accuracy, $\kappa = .24$), style convergence was non-significant, $F(1.90, 60.95) = 1.08$, $p = .342$, $\eta_p^2 = .03$, though few-shot prompting calibrated length (15.1–16.3 vs. 27.6 words, $F = 34.36$, $p < .001$; Figure E2). Re-running on the **Refined**

**User Archetypes** with the *Blocks* router ($M$ = 5 user messages, 70.1% raw agreement, $\kappa$ = .535) **rejected $H_{02}$ ($p$ < .001) on structural style and pacing calibration**: de-lexicalized SBERT style convergence differed significantly under Greenhouse–Geisser correction (Greenhouse & Geisser, 1959), $F(1.88, 60.29) = 9.79$, $p < .001$, $\eta_p^2 = .23$, and Friedman testing, $\chi^2(2) = 12.50$, $p = .002$. Cluster-Adaptive Few-Shot ($M$ = .597, $SD$ = .086) significantly outperformed Static Few-Shot ($M$ = .453, $t(32) = 4.24$, $p_{Holm} < .001$, $d_z = 0.74$) and Zero-Shot ($M$ = .530, $t(32) = 2.36$, $p_{Holm}$ = .049, $d_z = 0.41$), and cut archetype pacing error from 10.12 (Zero-Shot) and 6.33 (Static) to **3.87 words** ($F = 7.10$, $p = .007$; Friedman $\chi^2(2) = 14.27$, $p < .001$; Holm $p = .004$ vs. Zero-Shot, $p$ = .024 vs. Static). By contrast, generic semantic similarity metrics—including Gemini 3,072-d cosine (.375 vs. .348, $F = 0.61$, $p = .534$) and token-level BERTScore $F_1$ (.415 vs. .391, $F = 1.16$, $p = .315$; Table E3)—did not differ significantly across conditions, confirming that archetype-adaptive prompting selectively calibrates discourse structure and turn pacing without distorting underlying domain semantics.

**Table 14**

*Repeated-Measures Comparison of Prompting Strategies on Primary Versus Refined User Archetypes (RQ2)*

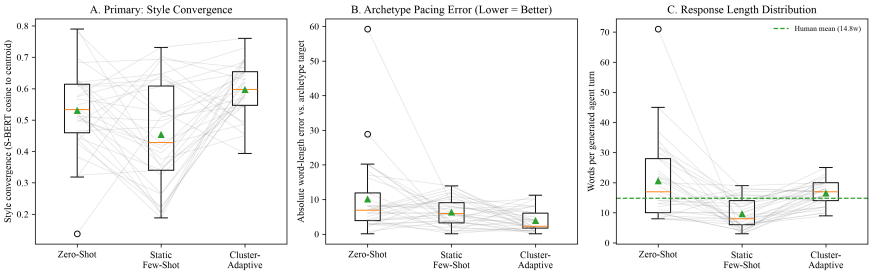
| Experiment and outcome metric | Zero-Shot *M* (*SD*) | Static Few-Shot *M* (*SD*) | Cluster-Adaptive *M* (*SD*) | RM-ANOVA *F* (GG) | *p* (GG) | $\eta_p^2$ |
|---|---|---|---|---|---|---|
| Primary archetypes (*M*=3 prefix, κ=.24) — Style cosine | .519 (.126) | .491 (.109) | .486 (.099) | 1.08 | .342 | .03 |
| Primary archetypes — Response length (words) | 25.2 (11.8) | 10.6 (7.0) | 12.3 (8.2) | 34.36 | < .001 | .52 |
| Refined archetypes (*M*=5 Blocks, κ=.535) — SBERT style cosine | .530 (.132) | .453 (.169) | .597 (.086) | 9.79 | < .001 | .23 |
| Refined archetypes — Gemini 3,072-d style cosine | .348 (.133) | .372 (.127) | .375 (.105) | 0.61 | .534 | .02 |
| Refined archetypes — Token-level BERTScore $F_1$ | .391 (.204) | .360 (.245) | .415 (.163) | 1.16 | .315 | .04 |

| Experiment and outcome metric | Zero-Shot *M* (*SD*) | Static Few-Shot *M* (*SD*) | Cluster-Adaptive *M* (*SD*) | RM-ANOVA *F* (GG) | *p* (GG) | $\eta_p^2$ |
|---|---|---|---|---|---|---|
| Refined archetypes — Pacing calibration error (words) | 10.12 (11.17) | 6.33 (4.14) | 3.87 (3.23) | 7.10 | .007 | .18 |
| Refined archetypes — Response length (words) | 20.6 (13.3) | 9.7 (5.0) | 16.5 (4.0) | 14.72 | < .001 | .32 |

*Note.* $N = 33$ held-out dialogues per experiment (99 generations each; *gemini-2.5-flash*). Cluster-Adaptive combines archetype-partitioned FAISS exemplars with an explicit archetype pacing directive. GG = Greenhouse–Geisser correction.

**Figure 6**

*Style Convergence and Archetype Pacing Calibration Across Prompting Conditions on Refined User Archetypes*




*Note.* $N = 33$ held-out dialogues. Panel A: De-lexicalized SBERT style convergence cosine. Panel B: Absolute pacing error (words).

**RQ3 Results: 2 × 2 Factorial Evaluation of Declarative Rules and Adaptive Prompting**

Table 15 and Figure 7 present the 2 × 2 factorial experiment across $N = 144$ evaluations (36 stratified held-out dialogues—18 policy-violating and 18 compliant—evaluated across all four conditions, $n = 36$/cell), where the *Rules Active* condition represents an integrated three-component governance package (pre-generation declarative YAML rule checking, [SLOT] exemplar masking, and post-generation regex leakage sanitization). Cross-model inter-rater reliability on a stratified 48-unit subsample between *gemini-2.5-flash* and independent *gemini-*

*2.5-pro* reached 95.8% raw agreement and Cohen's κ = .917 (PABAK = .917; Table C3). Four findings **reject $H_{03}$ ($p < .001$)**: (1) separating **Rule Adherence** into its two constituent scenario classes reveals that **Violation Enforcement Rate (Sensitivity, $n$ = 18 violating cases/cell)** rose from 5.6% (*SD* = 23.6%) under both *Rules Off* cells to **100.0%** (*Rules Active* × *Zero-Shot*) and **97.2%** (*SD* = 11.8%, *Rules Active* × *Cluster-Adaptive*), yielding a massive rules main effect, $F(1, 68) = 498.78$, $p < .001$, $\eta_p^2 = .880$, whereas **Compliant-Case Acceptance Rate (Specificity, 1 − false_block, $n$ = 18 compliant cases/cell)** remained statistically indistinguishable between *Rules Off* (94.4%, 17/18) and *Rules Active* (88.9%, 16/18), $F(1, 68) = 0.69$, $p = .408$, $\eta_p^2 = .010$, confirming that strict policy enforcement does not induce significant false-positive blocking; pooled across all 36 cases per cell, **Overall Rule-Decision Accuracy** rose from .500 to .944/.931, rules main effect $F(1, 140) = 43.92$, $p < .001$, $\eta_p^2 = .24$ (HC3 $F = 42.71$; ART $F = 39.12$, both $p < .001$), with zero prompting main effect ($F = 0.01$, $p = .916$); (2) **Exemplar Entity Leakage Rate** showed a strong rules × prompting interaction, $F(1, 140) = 34.82$, $p < .001$, $\eta_p^2 = .20$ (Sum-coded HC3 $F = 33.85$; ART $F = 2{,}148.81$, both $p < .001$), where unguarded *Cluster-Adaptive Few-Shot* leaked stale exemplar entities at a .359 rate (20/36 = 55.6% incidence) but *Rules Active* × *Cluster-Adaptive* reduced detected leakage in the evaluated sample to **0.0% (.000, 0/36 incidence; simple-effect Welch $t(35) = 5.90$, $p < .001$, Mann–Whitney $U = 1008.0$, $p < .001$, Fisher's exact $p = 4.68 \times 10^{-8}$)**; (3) **Slot Coverage $F_1$** rose from .208/.485 to **.977/.952** under active rules, $F(1, 140) = 489.52$, $p < .001$, $\eta_p^2 = .78$; and (4) **Style Alignment** was significantly boosted by Cluster-Adaptive Few-Shot, $F(1, 140) = 24.00$, $p < .001$, $\eta_p^2 = .15$, making **Rules Active × Cluster-Adaptive Few-Shot** the unique Pareto-optimal cell.

**Table 15**

*Cell Means and 2 × 2 Factorial ANOVA Results for RQ3 Outcomes (N = 144)*

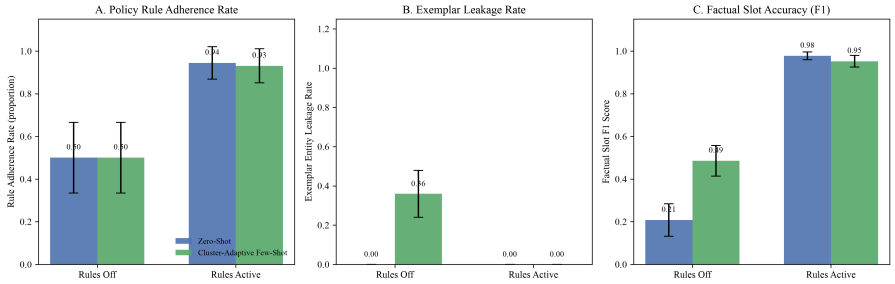
| Experimental cell or ANOVA effect | *n* or *df* | Rule Acc. *M* (*SD*) \| Sens. / Spec. | Exemplar leakage *M* (*SD*) / Inc. % | Slot coverage $F_1$ *M* (*SD*) | Style alignment *M* (*SD*) |
|---|---|---|---|---|---|
| Rules Off × Zero-Shot | 36 | .500 (.507) \| 5.6% / 94.4% | .000 (.000) / 0.0% | .208 (.233) | .412 (.086) |
| Rules Off × Cluster-Adaptive | 36 | .500 (.507) \| 5.6% / 94.4% | .359 (.365) / 55.6% | .485 (.219) | .521 (.088) |
| Rules Active × Zero-Shot | 36 | .944 (.232) \| 100.0% / 88.9% | .000 (.000) / 0.0% | .977 (.055) | .480 (.085) |
| Rules Active × Cluster-Adaptive | 36 | .931 (.244) \| 97.2% / 88.9% | .000 (.000) / 0.0% | .952 (.084) | .506 (.071) |
| ANOVA Main Effect: Rules Mode | 1, 140 | $F$=43.92 (< .001), $\eta_p^2$=.24 | $F$=34.82 (< .001), $\eta_p^2$=.20 | $F$=489.52 (< .001), $\eta_p^2$=.78 | $F$=3.68 (.057), $\eta_p^2$=.03 |
| ANOVA Main Effect: Prompting Mode | 1, 140 | $F$=0.01 (.916), $\eta_p^2$=.00 | $F$=34.82 (< .001), $\eta_p^2$=.20 | $F$=20.46 (< .001), $\eta_p^2$=.13 | $F$=24.00 (< .001), $\eta_p^2$=.15 |
| ANOVA Interaction: Rules × Prompting | 1, 140 | $F$=0.01 (.916), $\eta_p^2$=.00 | $F$=34.82 (< .001), $\eta_p^2$=.20 | $F$=29.32 (< .001), $\eta_p^2$=.17 | $F$=8.95 (.003), $\eta_p^2$=.06 |

*Note.* Rule Acc. = overall rule-decision accuracy ($n$ = 36/cell); Sens. = violation enforcement sensitivity on violating scenarios ($n$ = 18/cell; Rules $F(1, 68) = 498.78$, $p < .001$, $\eta_p^2 = .880$); Spec. = compliant-case acceptance specificity (1 − false_block, $n$ = 18/cell; Rules $F(1, 68) = 0.69$, $p = .408$); Inc. % = responses with ≥ 1 leaked entity. Dual-judge κ = .917. For exemplar leakage, because three of the four balanced 2 × 2 cells ($n$ = 36/cell) are identically .000 and only *Rules Off* × *Cluster-Adaptive* is non-zero ($M$ = .35926), all three orthogonal +1/−1 Sum-coded contrasts $c_A$ = (+1,+1,−1,−1), $c_B$ = (−1,+1,−1,+1), and $c_{AB}$ = (+1,−1,−1,+1) have identical contrast sum of squares $SS = n(\Sigma c_j \bar{Y}_j)^2 / \Sigma c_j^2 = 36(.35926)^2 / 4 = 1.162$ (equivalently, while the difference-of-differences interaction estimate −.359 is twice the marginal main-effect difference ±.180, its standard error $\sqrt{(4\sigma^2/n)}$ is also twice the marginal standard error $\sqrt{(\sigma^2/n)}$, yielding identical $t$ = ±5.901, OLS $F(1, 140) = t^2 = 34.82$, $\eta_p^2 = .199$, and Sum-coded HC3 $F = 33.85$, $p < .001$), whereas rank-based ART separates the main effects ($F$ = 43.48) from the interaction (ART $F$ =

2148.81, $p < .001$; simple-effect contrast within Cluster-Adaptive: Welch $t(35) = 5.90$, $p < .001$, Fisher's exact $p = 4.68 \times 10^{-8}$).

**Figure 7**

*2 × 2 Factorial Interaction Profiles for Rule Adherence, Exemplar Leakage, Slot Coverage F1, and Style Alignment*




*Note.* $N = 144$ held-out units ($n = 36$ per cell). Error bars represent 95% confidence intervals.

**RQ4 Results: 2 × 3 Multi-Turn Simulation of Turn Economy Across Archetypes**

Table 16 and Figure 8 report the 2 × 3 between-subjects multi-turn simulation across ***N* = 180 unique, independent held-out seed dialogues** (60 per archetype, partitioned into $n = 30$ distinct dialogues for *Zero-Shot* and $n = 30$ distinct dialogues for *Cluster-Adaptive Few-Shot* with zero overlap; human corpus baseline is in Appendix E, Table E5 and Figure E3). In this experiment, *total_turns* counts total conversational turns (user plus assistant messages) until the simulated traveler issues <code>[ACCEPT_BOOKING]</code>, and *Cluster-Adaptive Few-Shot* evaluates the combined package of the **Verified Informed Slot Ledger** (forbidding re-asking any previously stated slot) and **archetype-specific turn-economy policies**. Blind evaluation of 45 simulated transcripts by *gemini-2.5-pro* confirmed 95.6% within-simulation persona fidelity (43/45 vs. 33.3% chance, binomial $p < .001$, $\kappa = .933$; Table C3). On log-transformed turns to goal completion, the 2 × 3 between-subjects ANOVA **rejected $H_{04}$ ($p < .001$)**, with significant main effects of prompting, $F(1, 174) = 664.43$, $p < .001$, $\eta_p^2 = .79$, and

archetype, $F(2, 174) = 5.55$, $p = .005$, $\eta_p^2 = .06$, plus a **significant prompting × archetype interaction**, $F(2, 174) = 10.89$, $p < .001$, $\eta_p^2 = .11$ (HC3 $F = 8.58$; ART $F = 209.46$, both $p < .001$). Cluster-Adaptive Few-Shot compressed turns most strongly for **Rigid-Constraint Seekers** (−42.7%, 8.73 → 5.00 turns, Welch $t_{log} = -16.14$, $p < .001$, $d = -3.40$) by proactively presenting both 4-star exact-date (Option A) and 5-star flexible-date (Option D) trade-offs when the exact-date 5-star option (Option C) exceeded budget, followed by **Decisive Bookers** (−34.2%, 7.60 → 5.00 turns, $t_{log} = -19.26$, $d = -3.94$) and **Open Explorers** (−29.5%, 7.47 → 5.27 turns, $t_{log} = -11.24$, $d = -2.55$). Notably, the $M = 5.00$ ($SD = 0.00$) result for *Decisive Bookers* and *Rigid-Constraint Seekers* under *Cluster-Adaptive Few-Shot* reflects a structural protocol floor effect: because every session requires Turn 1 (user states core trip slots), Turn 2 (assistant asks for the single remaining unknown slot, budget), Turn 3 (user states budget), Turn 4 (assistant presents tailored package offer[s]), and Turn 5 (user issues <code>[ACCEPT_BOOKING]</code>, guarded by <code>ex_idx ≥ 1</code> and budget confirmation in *step4_rq4_simulation.py*), 5 total conversational turns is the theoretical minimum completion length, which 30/30 episodes in those two cells achieved on the first offer turn (motivating our paired HC3, ART, and Mann–Whitney $U$ checks). Redundant Elicitation Rate dropped from .194 (DB), .061 (OE), and .072 (RCS) in Zero-Shot to **0.000** across all archetypes in Cluster-Adaptive Few-Shot, main effect $F(1, 174) = 63.02$, $p < .001$, interaction $F(2, 174) = 9.70$, $p < .001$.

**Table 16**

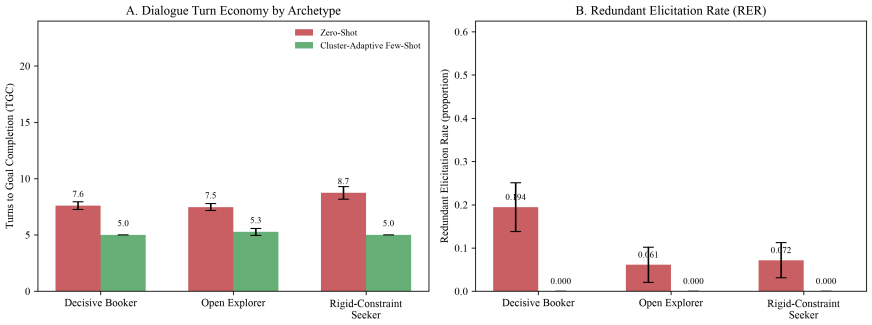
*Cell Means, Archetype Contrasts, and 2 × 3 Factorial ANOVA Results for Multi-Turn Turn Economy (N = 180)*

| Archetype × Prompting cell or ANOVA effect | Total turns *M* (*SD*) | Log turns *M* (*SD*) | RER *M* (*SD*) | Turn reduction / HC3 check | Welch contrast / ART check |
|---|---|---|---|---|---|
| Decisive Booker — Zero-Shot (*n* = 30) | 7.60 (0.93) | 2.021 (0.117) | .194 (.158) | — | — |
| Decisive Booker — Cluster-Adaptive (*n* = 30) | 5.00 (0.00) | 1.609 (0.000) | .000 (.000) | −2.60 (−34.2%) | *t*=−19.26 (< .001), *d*=−3.94 |
| Open Explorer — Zero-Shot (*n* = 30) | 7.47 (0.86) | 2.005 (0.108) | .061 (.114) | — | — |
| Open Explorer — Cluster-Adaptive (*n* = 30) | 5.27 (0.87) | 1.651 (0.134) | .000 (.000) | −2.20 (−29.5%) | *t*=−11.24 (< .001), *d*=−2.55 |
| Rigid-Constraint Seeker — Zero-Shot (*n* = 30) | 8.73 (1.55) | 2.151 (0.184) | .072 (.114) | — | — |
| Rigid-Constraint Seeker — Cluster-Adaptive (*n* = 30) | 5.00 (0.00) | 1.609 (0.000) | .000 (.000) | −3.73 (−42.7%) | *t*=−16.14 (< .001), *d*=−3.40 |
| ANOVA Main Effect: Prompting Mode (*df* = 1, 174) | *F*=457.74 (< .001), $\eta_p^2$=.72 | *F*=664.43 (< .001), $\eta_p^2$=.79 | *F*=63.02 (< .001), $\eta_p^2$=.27 | HC3 $F_{log}$=715.48 (< .001) | ART $F_{log}$=671.25 (< .001) |
| ANOVA Main Effect: Refined Archetype (*df* = 2, 174) | *F*=7.24 (< .001), $\eta_p^2$=.08 | *F*=5.55 (.005), $\eta_p^2$=.06 | *F*=9.70 (< .001), $\eta_p^2$=.10 | HC3 $F_{log}$=0.01 (.923) | ART $F_{log}$=9.13 (< .001) |
| ANOVA Interaction: Prompting × Archetype (*df* = 2, 174) | *F*=11.93 (< .001), $\eta_p^2$=.12 | *F*=10.89 (< .001), $\eta_p^2$=.11 | *F*=9.70 (< .001), $\eta_p^2$=.10 | HC3 $F_{log}$=8.58 (< .001) | ART $F_{log}$=209.46 (< .001) |

*Note.* *N* = 180 distinct held-out seed dialogues (*n* = 30 unique dialogues/cell; 100.0% goal completion). Total turns = user + assistant messages (protocol minimum = 5.00 turns: U1 → A1 → U2 → A2 → U3). Within-simulation persona fidelity audit (*gemini-2.5-pro*, *n* = 45): 95.6% accuracy, κ = .933.

**Figure 8**

*Dialogue Turn Economy and Redundant Elicitation Rate by User Archetype and Prompting Strategy*




*Note.* $N = 180$ simulations ($n = 30$/cell). Panel A: Mean turns to goal completion. Panel B: Mean Redundant Elicitation Rate (RER).

**Technical Interpretation, Business Significance, and Synthesis of Conclusions**

*Technical synthesis across RQ1–RQ4.* Together, the four experiments form an integrated four-stage research progression. First, RQ1 and RQ2 demonstrate that **in-context style adaptation is bounded by upstream representation and routing quality**: whole-dialogue primary archetypes suffered from outcome confounding and near-chance early routing (39.4%, $\kappa = .24$), yielding a null RQ2 style effect ($p = .211$); truncating to opening user behavior via the 46-d *Blocks* representation (bootstrap ARI = .89/.87) and routing in that exact space (70.1% raw agreement, $\kappa = .535$ at $m = 5$; 97.3% same-feature boundary consistency, $\kappa = .958$ at $m = 9$) unlocked significant structural style convergence on de-lexicalized SBERT cosine ($p < .001$, $\eta_p^2 = .234$) and reduced pacing error from 10.12 to 3.87 words ($p = .007$) while preserving semantic content (Gemini cosine and BERTScore $F_1$ $p \geq .315$). Second, RQ3 demonstrates **functional orthogonality with protective interaction**: adaptive prompting governs discourse style ($F = 24.00$, $p < .001$), whereas the declarative governance layer governs policy violation enforcement sensitivity (5.6% → 98.6%, $F = 498.78$, $p < .001$; overall accuracy $F = 43.92$, $p < .001$) and slot

grounding ($F = 489.52$, $p < .001$) while reducing the 35.9% entity leakage (55.6% incidence) of unguarded few-shot exemplars to zero detected leakage in the evaluated sample. Third, RQ4 demonstrates across 180 independent multi-turn simulations that pairing archetype-tailored recommendation policies with a Verified Informed Slot Ledger compresses total conversational turns by 29.5%–42.7% and eliminates redundant slot questions.

*Practical business significance (illustrative operational projection).* As an illustrative operational scenario, if an online travel agency or hospitality contact center handling 1,000,000 monthly sessions achieved a comparable turn reduction to that observed in the RQ4 simulation—reducing mean conversation length from 7.93 total turns (Zero-Shot) to 5.09 total turns (Cluster-Adaptive), a 35.8% reduction (2.84 turns per session)—it would save approximately 2.84 million message-generation calls per month, lowering inference compute and interaction latency by roughly one-third. Simultaneously, reducing redundant slot questions (10.9% → 0.0% in simulation), preventing exemplar price/promo leakage (55.6% incidence → 0.0% detected), and raising policy violation enforcement sensitivity from 5.6% to 98.6% (while maintaining 88.9% compliant-case acceptance) directly targets key drivers of customer drop-off and compliance risk.

## Implementation, Limitations, and Future Scope

### Deployment Architecture, System Integration, and Real-World Use Case

*Enterprise deployment and integration.* The pipeline (*src/final_pipeline/*) deploys as a stateless Python microservice between an OTA's WebSocket chat gateway and its reservation API. Active sessions store the canonical *DialogueTrajectory* and *Verified Informed Slot Ledger* in Redis, query in-memory *faiss-cpu* archetype sub-indices (< 2 ms), run deterministic YAML

policy checks via *src/rule_engine.py* (< 1 ms), and call *gemini-2.5-flash* with post-generation entity verification.

*Real-world use case.* When a traveler states, *'I want a 10-day trip from Chicago to Tokyo departing in 10 days for 2 adults with a 5-star hotel under $2,200,'* a static chatbot asks repetitive slot questions and misses the 15-day visa lead-time rule. Under our deployed system: (1) *RefinedArchetypeRouter* routes the session to *Rigid-Constraint Seeker*; (2) the slot ledger locks all five stated slots against re-asking; (3) *RuleEngine* flags *RULE_VISA_LEAD_TIME* (< 15 days) and detects that the exact-date 5-star package ($2,596) exceeds $2,200; and (4) the assistant immediately notifies the user of the 15-day visa requirement while presenting both the 4-star package ($1,936) and the flexible-date 5-star package ($2,068) within budget, completing the booking in 5 turns with zero compliance risk.

**Computational Reproducibility and Operational Requirements**

All code, random seeds (seed = 42), prompts, and JSONL caches (*data/interim/*.jsonl*, *data/final/*.jsonl*) are versioned at https://github.com/surnella/QM640-Dialogue-Clustering-Capstone. Re-running the entire study from cached responses on a standard laptop (Apple M-series or 4-core CPU, 16 GB RAM, Python 3.11) completes in under 4 minutes at zero API cost. Cold regeneration of all ~8,900 Gemini API calls across RQ1–RQ4 completes in ~35 minutes using 8–12 workers and costs under $4.50 USD at standard Gemini 2.5 Flash/Pro pricing.

**Study Limitations and Threats to Validity**

Five limitations bound this study: (1) **continuous archetype geometry and unsupervised reference labels**, as modest *K*-Means Silhouette scores (.094 on 46-d *Blocks*; .147–.161 across MiniLM and Gemini embeddings) indicate that traveler styles form a continuous behavioral manifold rather than discrete islands, and prefix routing κ (.535 at $m = 5$;

.958 same-feature boundary consistency at $m = 9$) is evaluated against unsupervised $M = 9$ *K*-Means reference clusters rather than external human persona labels; (2) **early-turn routing latency**, since moderate-to-high *Blocks* routing requires 5–9 user messages, leaving Turn 1 cold-start routing noisy; (3) **bundled multi-component treatments in RQ2, RQ3, and RQ4**, since RQ2 evaluated *Cluster-Adaptive Few-Shot* as a joint package (archetype-partitioned FAISS exemplars + routed archetype pacing directive), RQ3 evaluated *Rules Active* as a joint governance package (declarative *RuleEngine* + [SLOT] masking + post-generation regex sanitization), and RQ4 evaluated the *Verified Informed Slot Ledger* jointly with archetype-specific turn-economy policies; (4) **corpus and synthetic Tier-2 scope**, because Frames uses role-players and Tier-2 rules (visa, passport, discount caps) required synthetic injection in RQ3; and (5) **simulated users and protocol floor in RQ4**, where cooperative LLM user simulators (validated for within-simulation persona fidelity at 95.6%, $\kappa = .933$, rather than live human traffic) and the 5-turn minimum protocol boundary ($M = 5.00$, $SD = 0.00$ for two cells) may overstate turn reductions relative to noisy production sessions.

**Future Scope and Research Extensions**

Future doctoral research on *Dialogue Clustering and Methodologies to be Adapted for Human-Like Conversations* should pursue six extensions: (1) **orthogonal factorial ablations across RQ2, RQ3, and RQ4**—first, expanding the 3-condition RQ2 design into a 5-condition experiment (**Condition A**: Examples *None*, Directive *Off*; **Condition B**: Examples *Global/Static*, Directive *Off*; **Condition C**: Examples *Archetype-Partitioned*, Directive *Off*; **Condition D**: Examples *Global/Static*, Directive *On*; and **Condition E**: Examples *Archetype-Partitioned*, Directive *On*) at $N \geq 60$ dialogues to orthogonally isolate explicit pacing directives from sub-index exemplars (preliminary $N = 33$ diagnostic testing shows that when directives are *Off*,

*Condition C* improves style convergence over *Condition B* from .453 to .511, $p = .028$, and calibrates length to 15.03 words vs. the 14.85-word human benchmark, whereas when directives are *On*, the explicit routed directive dominates single-turn style at .597–.610 while retrieved exemplars tighten pacing variance); second, ablating pre-generation [SLOT] masking versus post-generation regex sanitization in RQ3; and third, crossing the *Verified Informed Slot Ledger* (Off/On) with *Archetype Action Policy* (Generic/Adaptive) in a 2 × 2 × 3 RQ4 design to separate ledger memory gains from persona policy gains; (2) **hierarchical context × behavior clustering and influencing-factor attribution**, developing a two-tier taxonomy ($C \times K$ = 24–36 micro-clusters over $N > 50{,}000$ enterprise dialogues) that separates macro situational trip regimes (party composition, budget stakes, travel lead window, and destination familiarity) from within-regime behavioral negotiation personas while avoiding the sample starvation and 35.9% entity leakage of naive 500-bucket parameter clustering; (3) **soft-mixture manifold routing** that blends exemplars using posterior probabilities for boundary travelers; (4) **turn-by-turn Bayesian persona tracking** from Turn 1 onward to capture mid-session persona drift; (5) **two-stage quotation-to-conversion funnel modeling** that decouples Stage 1 in-chat quotation agreement from Stage 2 post-quote purchase conversion under exogenous competitor and lodging controls; and (6) **constrained neuro-symbolic reinforcement learning and live OTA A/B testing** with human travelers and human domain raters across multilingual settings.

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# Appendix A

## Repository Structure, Replication Guide, and Raw Source Schema

All code, data schemas, cached LLM outputs, and report builders are available at https://github.com/surnella/QM640-Dialogue-Clustering-Capstone. Table A1 outlines the repository directory structure, Table A2 lists the exact shell commands that reproduce every table and figure, and Table A3 provides Part A (raw source fields) and Part B (canonical schema fields) of the data dictionary.

**Table A1**

*Repository Structure and Artifact Inventory*

| Directory or file path | Contents and analytical purpose |
|---|---|
| data/raw/ | Downloaded Microsoft Frames (1,369 dialogues) and MultiWOZ 2.2 corpora (excluded from Git) |
| data/canonical/ | Normalized two-tier Pydantic dialogue trajectories (canonical_corpus.jsonl) |
| data/interim/ | Cleaned samples, partitions, embeddings, primary/refined archetypes, LLM annotations, and RQ1/RQ2 JSON summaries |
| data/final/ | Cached Gemini 2.5 Flash/Pro JSONL responses and scored CSV/JSON results for Refined RQ2, RQ3 (*N*=144), and RQ4 (*N*=180) |
| src/canonical_schema.py, src/ingestion.py | Pydantic two-tier data models and corpus parsers |
| src/rule_engine.py, rules/rules_enterprise.yaml | Deterministic declarative policy engine and YAML rule definitions |
| src/interim_pipeline/ | Reproducible scripts for Steps 0–4 and RQ1 sensitivity/truncation Steps 2b–2i |
| src/final_pipeline/ | Reproducible scripts for RefinedArchetypeRouter, Refined RQ2, RQ3 2×2 experiment, RQ4 2×3 simulation, and EDA/architecture figures |
| src/report/, src/build_final_report.py | APA 7 Word/PDF document generator with automated cross-reference and TOC pagination verification |
| notebooks/01_.. to 05_.. | Five executed Jupyter notebooks covering EDA, RQ1, RQ2, RQ3, and RQ4 |
| outputs/interim/, outputs/final/ | All CSV summary tables and 300-DPI PNG figures |
| final/ | Submission-ready QM_640_Final_Report_Sandeep_Surnella.docx and .pdf |

**Table A2**

*End-to-End Replication Commands*

| Pipeline stage | Command (run from repository root with .venv active) | Outputs produced |
|---|---|---|
| 0. Power analysis | python src/interim_pipeline/step0_sample_size.py | Table 8 |
| 1. Cleaning & univariate EDA | python src/interim_pipeline/step1_prepare_data.py | Tables 5, 6, E1; Figures 1, E1 |
| 2. Primary RQ1 clustering | python src/interim_pipeline/step2_rq1_clustering.py | Table 12; Tables D1–D4; Figures D1–D3 |
| 2b–2i. RQ1 sensitivity & refined | python src/interim_pipeline/step2g_rq1_truncation.py && python src/interim_pipeline/step2h_rq1_ablation.py && python src/interim_pipeline/step2i_rq1_refined_archetypes.py | Table 13; Tables C1–C2, D5–D6, E5; Figures D4, E3 |
| 3. Primary RQ2 experiment | python src/interim_pipeline/step3_rq2_pipeline.py --execute | Table 14 (primary rows); Figure E2 |
| 4. Rule & baseline groundwork | python src/interim_pipeline/step4_rq3_rq4_groundwork.py | Table 9; Table E4 |
| 5. Final harness & routing test | python src/final_pipeline/test_step1_harness.py | Table 13 routing note (t_rq1_hybrid_routing_diagnostics.csv) |
| 6. Refined RQ2 experiment | python src/final_pipeline/step2_rq2_refined.py --execute | Table 14 (refined rows); Table E3; Figure 6 |
| 7. RQ3 2×2 factorial experiment | python src/final_pipeline/step3_rq3_experiment.py --execute | Table 15; Table C3; Figure 7 |
| 8. RQ4 2×3 turn-economy sim. | python src/final_pipeline/step4_rq4_simulation.py --execute | Table 16; Table C3; Figure 8 |
| 9. Bivariate/multivariate EDA & arch. | python src/final_pipeline/step5_figures_and_eda.py | Table 7; Table E2; Figures 2, 3, 4, 5 |
| 10. Build Final Report (.docx/.pdf) | python src/build_final_report.py | final/QM_640_Final_Report_Sandeep_Surnella.docx & .pdf |

*Note.* All random seeds are fixed at 42. Cached JSONL responses allow offline execution without API calls.

**Table A3**

*Data Dictionary, Parts A and B: Raw Corpus Fields and Canonical Schema Mapping*

| Source / Layer | Field name | Data type (Unit) | Description and mapping role |
|---|---|---|---|
| Frames (Raw) | id, user_id, wizard_id | String (ID) | Unique dialogue ID and role-player IDs (11 users, 12 wizards) |
| Frames (Raw) | turns[].author, text, timestamp | String / Float (ms) | Speaker (user or wizard), raw utterance text, and Unix timestamp |

| Source / Layer | Field name | Data type (Unit) | Description and mapping role |
|---|---|---|---|
| Frames (Raw) | turns[].labels.acts[].name, args | List / Key–Value | Dialogue act tags (inform, request, offer, switch_frame) and slot–value pairs |
| Frames (Raw) | labels.wizardSurveyTaskSuccessful, userSurveyRating | Boolean / Float (1–5) | Wizard task completion flag and user satisfaction rating |
| MultiWOZ (Raw) | dialogue_id, services | String / List | Unique dialogue ID and active service domains (hotel, train, taxi, attraction) |
| MultiWOZ (Raw) | turns[].speaker, utterance, frames[].state | String / Dict | Speaker (USER or SYSTEM), utterance text, and turn-level slot state dictionary |
| Canonical (Tier 1) | origin_city, destination_city, accommodation_tier | String (City / Tier) | Normalized origin, destination, and hotel star rating or category |
| Canonical (Tier 1) | departure_date, return_date, duration_nights | String (Date) / Int (nights) | Normalized trip dates and length of stay in nights |
| Canonical (Tier 1) | traveler_count_adults, traveler_count_children, budget_amount | Int (persons) / Float (USD) | Party composition and numeric budget ceiling in US dollars |
| Canonical (Tier 2) | visa_processing_window_days, passport_expiry_months | Int (days / months) | Departure lead window (days) and remaining passport validity (months) |
| Canonical (Tier 2) | package_type, sharing_basis, discount_applied_percentage | String / Float (%) | Package class (FIT or Group), room occupancy, and promotional discount (%) |

# Appendix B

## Ten-Week Project Schedule and Milestone Completion Log

Table B1 documents the completed 10-week execution schedule aligned with QM640 course milestones.

**Table B1**

*Ten-Week Project Schedule and Milestone Completion Status*

| Week | QM640 course topic | Completed project tasks and deliverables | Status |
|---|---|---|---|
| 1 | Define research problem & topic | Business & technical problem statements finalized; GitHub repository initialized; Frames & MultiWOZ 2.2 ingested | Complete |
| 2 | Synopsis draft | Four research questions, hypotheses, and two-tier Pydantic schema defined | Complete |
| 3 | Literature search & data collection | 26 peer-reviewed business & technical sources synthesized; canonical corpus parsed ($N$ = 2,834 analytic) | Complete |
| 4 | Objectives & hypotheses | Data quality audit, univariate/bivariate/multivariate EDA, noncentral-$F$ power analysis, and primary RQ1 clustering | Complete |
| 5 | Research design & methodology | Interim report submitted; primary RQ2 experiment (99 calls) and RQ1 sensitivity/truncation grid (Steps 2b–2i) completed | Complete |
| 6 | Data analysis plan | Built RefinedArchetypeRouter (Blocks prefix router, $\kappa$ = .535–.958); repeated RQ2 on refined archetypes ($p$ < .001) | Complete |
| 7 | Data analysis plan execution | Executed RQ3 2 × 2 factorial experiment ($N$ = 144) and dual-LLM judge reliability audit (gemini-2.5-pro, $\kappa$ = .917) | Complete |
| 8 | Data analysis assessment | Executed RQ4 2 × 3 multi-turn simulation ($N$ = 180), persona fidelity validation (95.6%, $\kappa$ = .933), and HC3/ART checks | Complete |
| 9 | Refinement & documentation | Generated publication architecture, workflow, and EDA diagrams; synchronized empirical Jupyter notebooks and README | Complete |
| 10 | Document results & findings | Compiled, audited, and exported submission-ready Final Report (.docx and .pdf) | Complete |

# Appendix C

## LLM Behavioral Annotation Codebook, Reliability, and Dual-Judge Validation

Table C1 defines all 16 structured fields of the LLM behavioral codebook (`UserBehaviour` in *step2e_llm_user_annotation.py*: 10 ordinal 1–5 rating scales, 3 categorical goal/tone clustering fields, and 3 audit-only fields excluded from clustering), Table C2 reports test–retest reliability across all 16 fields and criterion validity against ground-truth Frames dialogue acts, and Table C3 summarizes the cross-model dual-judge reliability in RQ3 and simulator persona validation in RQ4.

**Table C1**

*LLM Behavioral Annotation Codebook (All 16 Structured Schema Fields)*

| # & Schema field | Scale & role | Operational definition and scale anchors |
|---|---|---|
| 1. destination_flexibility | Ordinal 1–5 (Cluster) | 1 = insists on one fixed destination; 5 = open to any destination recommendation |
| 2. date_flexibility | Ordinal 1–5 (Cluster) | 1 = fixed travel dates; 5 = wide date window or completely flexible timing |
| 3. exploration_breadth | Ordinal 1–5 (Cluster) | 1 = evaluates a single option; 5 = actively compares multiple destinations, hotels, or dates |
| 4. decisiveness | Ordinal 1–5 (Cluster) | 1 = hesitant or leaves undecided; 5 = commits quickly and firmly once a matching offer is shown |
| 5. detail_orientation | Ordinal 1–5 (Cluster) | 1 = asks no package details; 5 = probes amenities, star ratings, flight times, and inclusions |
| 6. revision_tendency | Ordinal 1–5 (Cluster) | 1 = never modifies initial constraints; 5 = repeatedly shifts dates, destinations, or budgets |
| 7. question_asking | Ordinal 1–5 (Cluster) | 1 = passive responder; 5 = asks frequent clarifying and comparison questions |
| 8. politeness | Ordinal 1–5 (Cluster) | 1 = curt or abrupt; 5 = consistently courteous and appreciative |
| 9. price_sensitivity | Ordinal 1–5 (Cluster) | 1 = price never mentioned or irrelevant; 5 = price/budget dominates every decision |
| 10. information_upfront | Ordinal 1–5 (Cluster) | 1 = reveals constraints gradually across turns; 5 = states nearly all trip constraints in opening message |
| 11. primary_goal | Categorical (Cluster) | 6 classes: explore_destinations, price_shopping, specific_trip_booking, compare_options, information_only, other |
| 12. secondary_goals | Multi-label (Cluster) | Multi-hot subset of the 6 goal classes above clearly present in user utterances |

| # & Schema field | Scale & role | Operational definition and scale anchors |
|---|---|---|
| 13. emotional_tone | Categorical (Cluster) | 4 classes: positive, neutral, impatient, or frustrated |
| 14. outcome | Categorical (Audit) | 4 classes: booked, no_suitable_option, undecided_or_left, unclear (excluded from clustering input) |
| 15. party_type | Categorical (Audit) | 6 classes: solo, couple, family, group, business, unknown (excluded from clustering input) |
| 16. unusual_conversation | Boolean (Audit) | True only if off-task or anomalous request (excluded from clustering input) |

*Note.* Fields 1–13 form the clustering feature matrix (*llm_matrix*: 10 ordinal + 6 primary_goal + 6 secondary_goals + 4 emotional_tone = 26 numeric columns). Fields 14–16 are retained strictly for post-clustering validation.

**Table C2**

*Test–Retest Reliability and Criterion Validity Across All 16 LLM Behavioral Schema Fields*

| Schema field (#1–16) | Test–retest κ | Ground-truth Frames act criterion or field role | Spearman ρ ($p < .001$) |
|---|---|---|---|
| 1. Destination flexibility | .96 | — | — |
| 2. Date flexibility | .89 | — | — |
| 3. Exploration breadth | .95 | Frame switches per user turn | .30 |
| | | Comparison requests per user turn | .26 |
| 4. Decisiveness | .91 | — | — |
| 5. Detail orientation | .89 | Distinct preference slots mentioned | .45 |
| 6. Revision tendency | .89 | Slot-value changes in user acts | .73 |
| 7. Question asking | .90 | Request acts per user turn | .65 |
| 8. Politeness | .91 | — | — |
| 9. Price sensitivity | .95 | Numeric budget stated | .28 |
| 10. Information given up front | .87 | Share of slots stated in first user turn | .32 |
| 11. Primary goal (6 classes) | .78 | Categorical Cohen's κ (clustering) | — |
| 12. Secondary goals (multi-label) | .75 | Binary presence Cohen's κ (clustering) | — |
| 13. Emotional tone (4 classes) | .88 | Categorical Cohen's κ (clustering) | — |
| 14. Session outcome (4 classes) | .98 | Categorical Cohen's κ (audit-only) | — |
| 15. Party type (6 classes) | .93 | Categorical Cohen's κ (audit-only) | — |
| 16. Unusual conversation (boolean) | .80 | Boolean Cohen's κ (99.0% raw agreement, audit-only) | — |

*Note.* Test–retest subsample $n = 100$ (quadratic-weighted κ for ordinal Fields 1–10; Cohen's κ for Fields 11–16); criterion validity $N = 1{,}334$ Frames dialogues.

**Table C3**

*Cross-Model Dual-Judge Reliability (RQ3) and Simulator Persona Fidelity Validation (RQ4)*

| Validation audit | Sample *n* | Evaluator models compared | Raw agreement / Accuracy | Cohen's κ (PABAK) | Additional validity evidence |
|---|---|---|---|---|---|
| RQ3 Policy Compliance Dual-Judge Audit | 48 | *gemini-2.5-flash* vs. *gemini-2.5-pro* | 95.8% | .917 (.917) | Flash vs. deterministic regex audit: 89.6% agreement, κ = .791 |
| RQ4 Simulator Persona Fidelity Audit | 45 | Intended Archetype vs. Blind *gemini-2.5-pro* Judge | 95.6% (vs. 33.3% chance) | .933 ($p$ < .001) | Mean ratings (Decisiveness / Exploration / Rigidity): DB = 5.0/1.0/1.5; OE = 3.9/4.3/2.0; RCS = 4.5/1.1/4.3 |

# Appendix D

## RQ1 Sensitivity Grid, Ablation Diagnostics, and Primary Archetype Profiles

Tables D1–D6 and Figures D1–D4 provide the complete diagnostic tables and visual plots supporting the RQ1 primary clustering benchmark and the representation × truncation sensitivity analysis.

### Table D1

*Choice of Number of Clusters k on the Golden Partition Across Initial MiniLM and Final 46-d Blocks (M = 9) Representations*

| Representation & *k* | Cosine Silhouette | Davies–Bouldin index | Inertia / Bootstrap ARI | Smallest cluster (%) & diagnostic |
|---|---|---|---|---|
| Panel A (MiniLM-L6-v2 whole): *k* = 2 | .206 | 3.136 | Inertia = 193.1 | 35.5% |
| Panel A (MiniLM-L6-v2 whole): *k* = 3 | .141 | 3.244 | Inertia = 184.6 | 24.1% |
| Panel A (MiniLM-L6-v2 whole): *k* = 4 | .142 | 2.997 | Inertia = 178.1 | 17.7% |
| Panel A (MiniLM-L6-v2 whole): *k* = 5 | .148 | 2.950 | Inertia = 173.6 | 8.7% |
| Panel A (MiniLM-L6-v2 whole): *k* = 6 | .077 | 3.118 | Inertia = 169.7 | 8.4% |
| Panel B (46-d Blocks, *M* = 9): *k* = 2 | .112 (cal. .044) | 2.796 | ARI = .949 (96.2% core) | 40.4% — Collapses Decisive & Rigid into 1 cluster |
| Panel B (46-d Blocks, *M* = 9): *k* = 3 | .094 (cal. .033) | 2.647 | ARI = .888 (93.1% core) | 25.5% — Adopted winner (lowest DB, ARI >= .60, min >= 20%) |
| Panel B (46-d Blocks, *M* = 9): *k* = 4 | .066 (cal. .015) | 2.756 | ARI = .589 (55.8% core) | 12.5% — Fails stability (ARI < .60) & min size (< 20%) |
| Panel B (46-d Blocks, *M* = 9): *k* = 5 | .069 (cal. .026) | 2.784 | ARI = .606 (61.1% core) | 11.4% — Fails stability (ARI < .60) & min size (< 20%) |
| Panel B (46-d Blocks, *M* = 9): *k* = 6 | .065 (cal. .019) | 2.729 | ARI = .631 (53.2% core) | 11.3% — Fragmented min cluster (11.3% < 20% floor) |

*Note.* On the final 46-d *Blocks* (*M* = 9) representation (Panel B), *k* = 3 achieves the lowest (best) Davies–Bouldin index (2.647), high bootstrap stability (ARI = .888 in the truncation grid; .869 in

the 6-config ablation), and balanced cluster sizes (≥ 25.5%), whereas $k = 2$ collapses *Decisive Booker* and *Rigid-Constraint Seeker* and $k \geq 4$ fails stability/size thresholds.

**Table D2**

*Pairwise Post Hoc Comparisons of Mean Silhouette Across Clustering Algorithms*

| Algorithm comparison | Mean difference | 95% CI (Tukey HSD) | *p* (Tukey HSD) | *p* (Games–Howell) |
|---|---|---|---|---|
| K-Means − Agglomerative | .029 | [.013, .046] | < .001 | < .001 |
| HDBSCAN − Agglomerative | .093 | [.076, .110] | < .001 | < .001 |
| HDBSCAN − K-Means | .063 | [.047, .080] | < .001 | < .001 |

**Table D3**

*Behavioral Profiles of the Three Primary Whole-Dialogue Archetypes (Golden Partition)*

| Feature | Direct Transactional | Constraint Negotiator | Guided Explorer |
|---|---|---|---|
| *n* (%) | 240 (25.7%) | 225 (24.1%) | 468 (50.2%) |
| Turns, *M* (*SD*) | 9.92 (5.02) | 15.64 (6.84) | 17.19 (6.10) |
| Words per user / agent turn | 10.60 / 11.67 | 12.94 / 18.15 | 9.63 / 15.71 |
| Slot-value revisions / Hesitations | 6.9 / 0.7 | 20.3 / 1.1 | 16.7 / 1.1 |
| Agent package offers | 0.84 | 4.28 | 3.73 |
| Budget stated (%) / Task success (%) | 63.7% / 92.1% | 70.7% / 94.2% | 75.4% / 95.3% |

**Table D4**

*Primary Archetype Assignment and Centroid Similarity Across Partitions and MultiWOZ 2.2*

| Sample | *N* | Centroid cosine *M* (*SD*) | Direct Transactional (%) | Constraint Negotiator (%) | Guided Explorer (%) |
|---|---|---|---|---|---|
| Frames golden | 933 | .895 (.042) | 25.7 | 24.1 | 50.2 |
| Frames held-out | 401 | .894 (.042) | 29.4 | 20.2 | 50.4 |
| MultiWOZ 2.2 | 1,499 | .805 (.037) | 0.9 | 15.1 | 84.0 |

**Table D5**

*Stability, Routing Agreement, and Outcome Association of Hybrid Archetypes by Conversation Length*

| User messages used (*M*) | Dialogues shortened (%) | Bootstrap ARI | Core members (%) | Smallest cluster (%) | Raw L1 κ | Blocks router κ | Outcome *V* |
|---|---|---|---|---|---|---|---|
| First 3 messages | 94.7 | .89 | 92.7 | 18.3 | .72 | .359 | .16 |
| First 5 messages | 78.2 | .84 | 77.1 | 20.3 | .38 | .535 | .27 |
| First 7 messages | 53.2 | .85 | 89.9 | 29.7 | .25 | .722 | .38 |
| First 9 messages | 30.7 | .89 | 93.1 | 25.5 | .24 | .958 | .45 |
| Whole conversation | 0.0 | .62 | 53.7 | 7.9 | .22 | — | .57 |

*Note.* Raw L1 κ uses unprojected 5-turn features (step2g); Blocks router κ projects prefix features into the exact golden Blocks space (RefinedArchetypeRouter).

**Table D6**

*Ablation of the Refined Archetype Definition at M = 9 User Messages (k = 3)*

| Representation configuration | ARI (whole) | ARI (9 msgs) | Core members (%) | Outcome *V* | ARI vs. adopted |
|---|---|---|---|---|---|
| Gemini embedding + Gemini behavior profile (adopted) | .56 | .87 | 89.2 | .45 | 1.00 |
| MiniLM embedding + Gemini behavior profile | .69 | .85 | 87.7 | .40 | .33 |
| Gemini behavior profile only | .73 | .85 | 97.6 | .35 | .17 |
| Gemini embedding only | .91 | .91 | 92.0 | .34 | .45 |
| MiniLM embedding only | .68 | .72 | 72.1 | .27 | .14 |
| MiniLM embedding + dialogue-act features (no LLM) | .61 | .76 | 88.0 | .45 | .16 |

**Figure D1**

*Cluster-Number Diagnostics on Golden Partition*

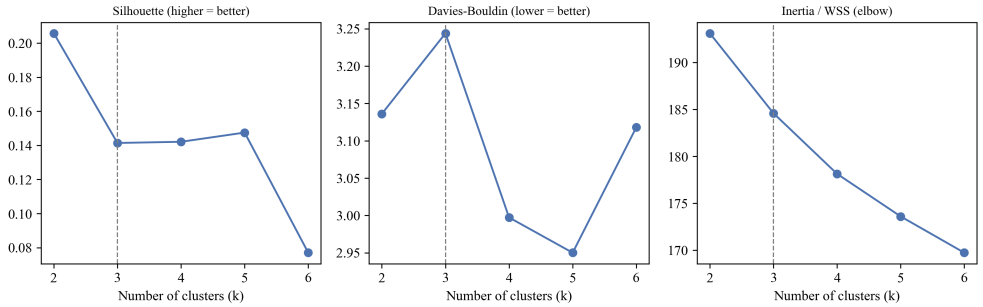



**Figure D2**

*Silhouette Coefficients Across 159 Subsample Runs by Algorithm*

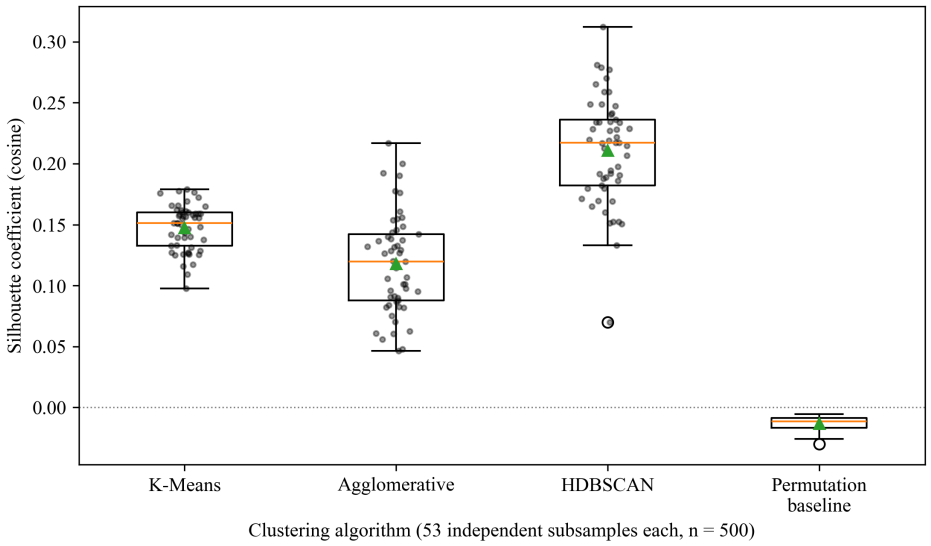



**Figure D3**

*UMAP Projection of Frames Archetypes and MultiWOZ 2.2 External Transfer Check*

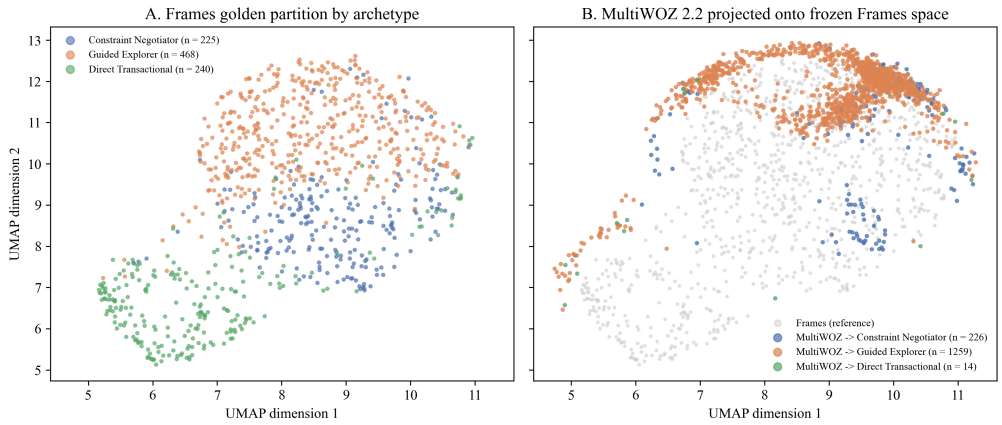
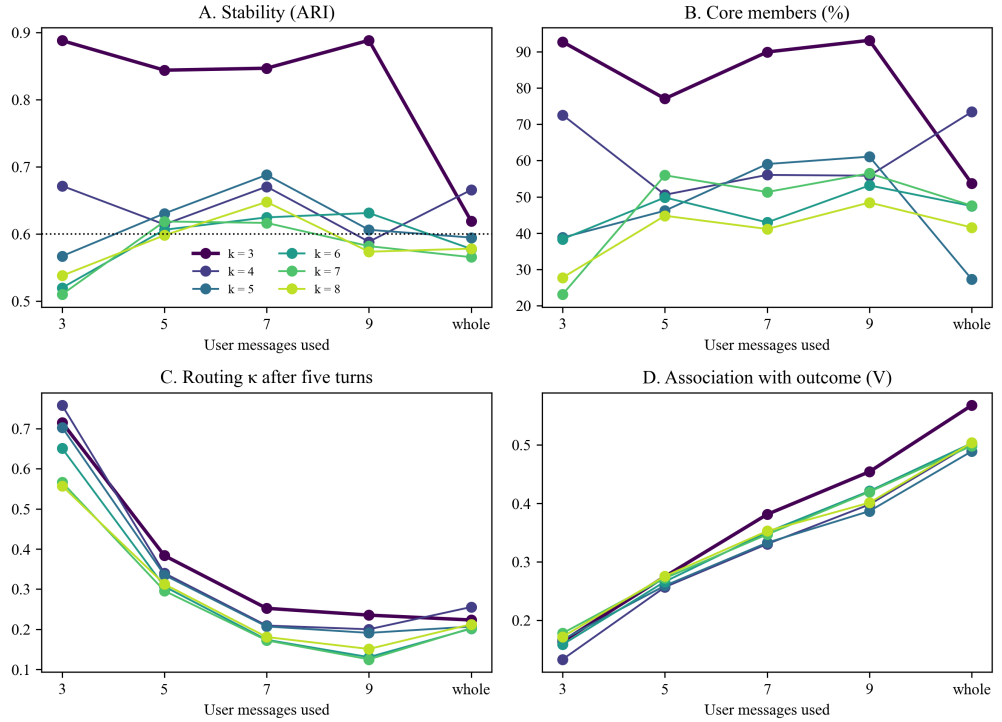



**Figure D4**

*Hybrid Archetype Stability, Routing, and Outcome Association Across Truncation Horizons*



# Appendix E

## Supplementary EDA, RQ2 Pairwise Contrasts, and RQ3/RQ4 Groundwork Diagnostics

Tables E1–E5 and Figures E1–E3 provide the supplementary slot-completeness table, multivariate PCA loadings, Holm-adjusted pairwise contrasts for RQ2, corpus rule coverage audit for RQ3, and human turn-economy baseline for RQ4.

### Table E1

*Observed Canonical Slot Completeness by Corpus*

| Canonical slot | Schema tier | Microsoft Frames (%) | MultiWOZ 2.2 (%) |
|---|---|---|---|
| origin city | Tier 1 | 99.4 | 70.8 |
| destination city | Tier 1 | 99.0 | 71.4 |
| departure date | Tier 1 | 84.9 | 68.6 |
| return date | Tier 1 | 70.8 | 0.0 |
| duration nights | Tier 1 | 61.8 | 27.0 |
| traveler count adults | Tier 1 | 63.9 | 54.0 |
| traveler count children | Tier 1 | 18.4 | 0.0 |
| budget amount | Tier 1 | 72.9 | 24.1 |
| accommodation tier | Tier 1 | 72.7 | 19.7 |
| visa processing window days | Tier 2 | 0.0 | 0.0 |
| passport expiry months | Tier 2 | 0.0 | 0.0 |
| discount applied percentage | Tier 2 | 0.0 | 0.0 |

### Table E2

*Multivariate Principal Component Loadings of Structural and Behavioral Features (Frames)*

| Variable | PC1 loading (40.8% var.) | PC2 loading (17.2% var.) | PC3 loading (10.6% var.) |
|---|---|---|---|
| Total Turns | .393 | .166 | −.234 |
| Slot Revisions | .424 | .195 | −.064 |
| Agent Offers | .361 | .292 | −.078 |
| User Words/Turn | .025 | −.087 | .888 |
| Dest. Flexibility | .327 | −.182 | .135 |
| Exploration Breadth | .417 | −.108 | .101 |
| Decisiveness | −.259 | .503 | .028 |
| Revision Tendency | .297 | −.460 | .025 |
| Question Asking | .301 | .248 | .027 |
| Politeness | .089 | .520 | .341 |

**Table E3**

*Holm-Adjusted Paired Contrasts for Refined RQ2 Outcomes (N = 33)*

| Outcome metric | Pairwise contrast | Mean diff. | *t* | *df* | *p* (Holm) | $d_z$ |
|---|---|---|---|---|---|---|
| Refined: SBERT style cosine | Cluster-Adaptive vs. Static | .144 | 4.24 | 32 | < .001 | 0.74 |
| Refined: SBERT style cosine | Cluster-Adaptive vs. Zero-Shot | .067 | 2.36 | 32 | .049 | 0.41 |
| Refined: SBERT style cosine | Static vs. Zero-Shot | −.077 | −2.20 | 32 | .049 | −0.38 |
| Refined: Gemini style cosine | Cluster-Adaptive vs. Zero-Shot | .027 | 1.20 | 32 | .713 | 0.21 |
| Refined: Gemini style cosine | Static vs. Zero-Shot | .024 | 0.79 | 32 | .865 | 0.14 |
| Refined: Gemini style cosine | Cluster-Adaptive vs. Static | .004 | 0.13 | 32 | .895 | 0.02 |
| Refined: BERTScore F1 | Cluster-Adaptive vs. Static | .056 | 1.45 | 32 | .472 | 0.25 |
| Refined: BERTScore F1 | Cluster-Adaptive vs. Zero-Shot | .025 | 0.85 | 32 | .804 | 0.15 |
| Refined: BERTScore F1 | Static vs. Zero-Shot | −.031 | −0.75 | 32 | .804 | −0.13 |
| Refined: Pacing error (words) | Cluster-Adaptive vs. Zero-Shot | −6.25 | −3.49 | 32 | .004 | −0.61 |
| Refined: Pacing error (words) | Cluster-Adaptive vs. Static | −2.46 | −2.66 | 32 | .024 | −0.46 |
| Refined: Pacing error (words) | Static vs. Zero-Shot | −3.79 | −1.82 | 32 | .078 | −0.32 |
| Refined: Response length (words) | Cluster-Adaptive vs. Static | 6.82 | 7.00 | 32 | < .001 | 1.22 |
| Refined: Response length (words) | Static vs. Zero-Shot | −10.91 | −4.29 | 32 | < .001 | −0.75 |
| Refined: Response length (words) | Cluster-Adaptive vs. Zero-Shot | −4.09 | −1.84 | 32 | .075 | −0.32 |

**Table E4**

*Empirical Corpus Audit of Policy Rule Coverage in Microsoft Frames (N = 1,334)*

| Declarative rule | Required canonical slot | Slot observed (%) | Dialogues triggering (*n*) | Triggering (%) |
|---|---|---|---|---|
| pax capacity | traveler count adults | 24.1 | 54 | 4.05 |
| budget minimum | budget amount | 65.1 | 266 | 19.94 |
| visa window | visa processing window days | 0.0 | 0 | 0.00 |
| passport validity | passport expiry months | 0.0 | 0 | 0.00 |
| discount threshold | discount applied percentage | 0.0 | 0 | 0.00 |

**Table E5**

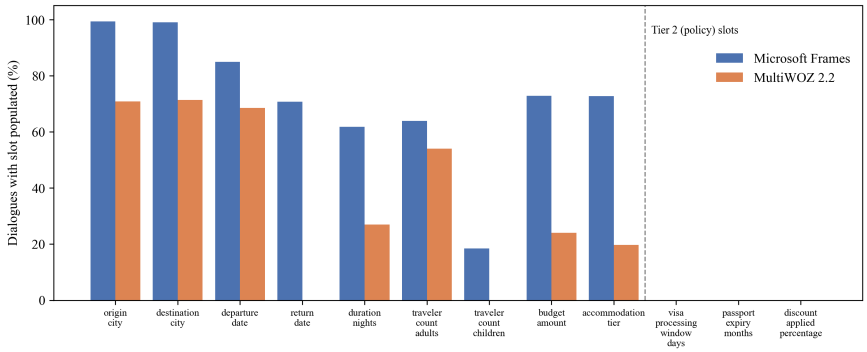
*Human Wizard Baseline of Turn Economy and Redundant Elicitation by Refined Archetype (Frames)*

| Refined user archetype | *n* | Whole-dialogue turns *M* (*SD*) | RER *M* (*SD*) | Wizard request turns *M* |
|---|---|---|---|---|
| Decisive Booker | 595 | 14.56 (4.52) | .014 (.047) | 1.44 |
| Open Explorer | 337 | 20.75 (7.07) | .016 (.049) | 1.82 |
| Rigid-Constraint Seeker | 402 | 10.60 (5.59) | .017 (.061) | 1.27 |

*Note.* Turns in Table E5 reflect total observed human-wizard turns (user + wizard messages) across complete archival Frames transcripts (including post-offer discussion and non-booking endings), whereas Table 16 (RQ4) measures total conversational turns until package agreement (<code>[ACCEPT_BOOKING]</code>) in closed-loop simulation.

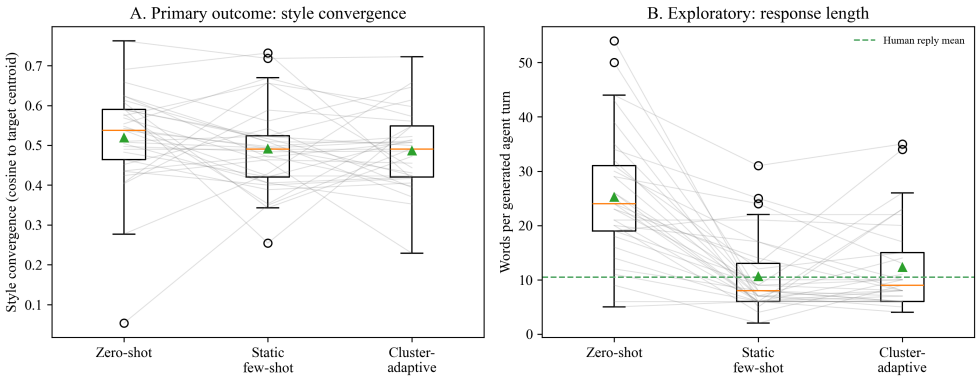
## Figure E1

*Observed Slot Completeness by Corpus*



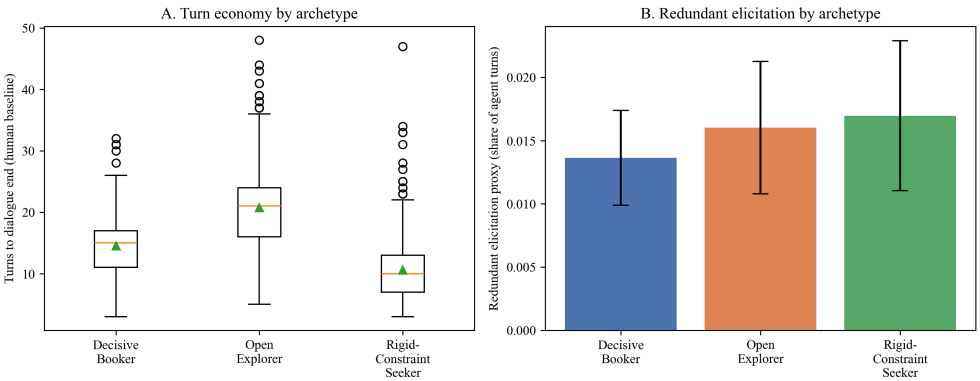

## Figure E2

*Primary RQ2 Results on Unrefined Whole-Dialogue Archetypes*




## Figure E3

*Human Wizard Baseline of Turn Economy and Redundant Elicitation by Refined Archetype*



# Appendix F

## Derivation of User Archetype Cluster Names and Deterministic Centroid-Mapping Protocol

Unsupervised clustering algorithms such as *K*-Means output anonymous integer cluster identifiers (<code>0</code>, <code>1</code>, <code>2</code>) rather than semantic persona names, and those integer identifiers are invariant up to permutation across random initializations (the *label-switching problem*). To ensure that human-interpretable archetype labels are assigned reproducibly without manual post-hoc cherry-picking, both the primary whole-dialogue clustering pipeline (*src/interim_pipeline/step2_rq1_clustering.py*) and the refined opening-prefix clustering pipeline (*src/interim_pipeline/step2i_rq1_refined_archetypes.py*) apply pre-declared, deterministic Python centroid-mapping rules to the golden-partition cluster means. Table F1 documents the exact sequential decision rule, distinguishing centroid criterion, and resulting golden-partition profile for each primary and refined archetype.

**Table F1**

*Deterministic Centroid-Mapping Rules for Primary and Refined User Archetypes*

| Clustering taxonomy & script | Step & deterministic Python rule on cluster centroid means | Assigned archetype name | Distinguishing golden-partition centroid signature ($N = 933$) |
|---|---|---|---|
| Primary Whole-Dialogue Archetypes (*step2_rq1_clustering.py*) | Step 1: <code>idxmax(turns_M)</code> among the 3 clusters | Guided Explorer | Longest dialogues ($M = 19.65$ turns, $n = 187$, 20.0%); highest option exploration across whole transcripts |
| Primary Whole-Dialogue Archetypes (*step2_rq1_clustering.py*) | Step 2: <code>idxmax(revisions)</code> among the remaining 2 clusters | Constraint Negotiator | Highest slot-revision count ($M = 1.98$ revisions, $M = 15.80$ turns, $n = 351$, 37.6%) |
| Primary Whole-Dialogue Archetypes (*step2_rq1_clustering.py*) | Step 3: Remaining cluster ID | Direct Transactional | Shortest whole-dialogue sessions ($M = 11.98$ turns, $n = 395$, 42.3%); lowest revision count |

| Clustering taxonomy & script | Step & deterministic Python rule on cluster centroid means | Assigned archetype name | Distinguishing golden-partition centroid signature ($N = 933$) |
|---|---|---|---|
| Refined Opening-Prefix Archetypes (*step2i_rq1_refined_archetypes.py*) | Step 1: <code>idxmax(destination_flexibility)</code> across the 3 *Blocks* clusters ($M = 9$) | Open Explorer | Highest destination flexibility (3.49/5.0) and exploration breadth (2.96/5.0); longest dialogues (20.78 turns; $n = 238$, 25.5%) |
| Refined Opening-Prefix Archetypes (*step2i_rq1_refined_archetypes.py*) | Step 2: <code>idxmax(decisiveness)</code> among the remaining 2 clusters | Decisive Booker | Highest decisiveness (3.97/5.0), politeness (4.26/5.0), and specific trip booking goal (99.3%; $n = 420$, 45.0%) |
| Refined Opening-Prefix Archetypes (*step2i_rq1_refined_archetypes.py*) | Step 3: Remaining cluster ID | Rigid-Constraint Seeker | Lowest destination/date flexibility (2.20/2.18) and detail orientation (1.46/5.0); highest non-booking rate under rigid constraints (68.0% no suitable option; $n = 275$, 29.5%) |

*Note.* All centroid-mapping rules are evaluated strictly on golden-partition means ($D_{\text{gold}}$, $n = 933$) after fitting $K$-Means ($k = 3$, seed = 42). Held-out test dialogues ($D_{\text{test}}$, $n = 401$) inherit the mapped archetype name of their nearest golden centroid.

# Appendix G

## Role, Architecture, and Validation Scope of Judge and Annotator LLMs

A critical methodological principle of this study is that **all primary hypothesis tests ($H_{01}$ through $H_{04}$) are anchored in deterministic mathematical and programmatic Python verification metrics** rather than subjective LLM preference scores. Specifically, RQ1 cluster quality is evaluated via mathematical Silhouette, Davies–Bouldin, Calinski–Harabasz, and Adjusted Rand Index formulas; RQ2 style convergence and pacing calibration are evaluated via deterministic non-Gemini encoders (*all-MiniLM-L6-v2* cosine, *roberta-large* BERTScore $F_1$, and exact word-count error); RQ3 exemplar entity leakage (ELR) and slot grounding $F_1$ are computed by deterministic Python string/set verifiers (*src/final_pipeline/metrics.py*); and RQ4 turn economy (turns to agreed package quotation and Redundant Elicitation Rate) is computed by deterministic turn counters and regex slot-ledger matching.

Within this architecture, LLMs serve two strictly bounded measurement and validation functions: (a) in RQ1, *gemini-2.5-flash* (at temperature = 0) acts as a **structured behavioral codebook annotator** prior to *K*-Means clustering, extracting the 16-field *UserBehaviour* schema (validated via test–retest $\kappa_w$ = .87–.96 and ground-truth Frames dialogue-act correlations $\rho$ = .30–.73; Table C2); and (b) in RQ3 and RQ4, the larger *gemini-2.5-pro* model serves strictly as an **independent cross-verification and quality-control validator** (Table C3). Importantly, the independent Judge LLM does not invent or contribute any standalone performance metric to firm up the paper's statistical conclusions; rather, it acts as an methodological gatekeeper. During the RQ3 pilot phase, cross-checking outputs against the deterministic verifier identified a prompt-ordering sensitivity (*pilot_v1* → *pilot_v2*) that was resolved prior to full execution. In the final experiments, because the independent *gemini-2.5-pro* validator strongly corroborated both the

RQ3 policy evaluations (95.8% agreement, κ = .917 with *gemini-2.5-flash*; 93.1% agreement, κ = .861 with the deterministic regex verifier) and the RQ4 simulated customer personas (95.6% blind classification accuracy, κ = .933), no manual re-annotation or rework was required. Table G1 summarizes the exact role, deterministic counterpart, and gatekeeping outcome of every LLM annotator and judge in the pipeline.

**Table G1**

*Functional Scope, Primary Deterministic Metrics, and Cross-Verification Role of Judge and Annotator LLMs Across RQ1–RQ4*

| Research question & pipeline stage | Primary hypothesis-test metrics (Deterministic / Mathematical) | LLM annotator / Judge model & role | Cross-verification agreement & gatekeeping outcome |
|---|---|---|---|
| RQ1: Opening-Prefix Behavioral Annotation (*step2e_llm_user_annotation.py*) | Cosine Silhouette, Davies–Bouldin, Calinski–Harabasz, Bootstrap ARI (.89 truncation grid / .87 ablation), and *Blocks* prefix-to-cluster κ (.359–.958) | **Codebook Annotator (*gemini-2.5-flash*, temp = 0):** Extracts Fields 1–13 (clustering inputs) and Fields 14–16 (audit-only) before *K*-Means runs; does not name clusters. | Test–retest $\kappa_w$ = .87–.96 (mean .91) on ordinal Fields 1–10; κ = .75–.98 on Fields 11–16; significant Spearman ρ = .30–.73 ($p < .001$) against ground-truth Frames acts (Table C2). |
| RQ2: Style Convergence & Pacing Calibration (*step2_rq2_refined.py*) | Hybrid & pure SBERT style cosine (*all-MiniLM-L6-v2*), BERTScore $F_1$ (*roberta-large*), and absolute word-count error ($\lvert Words_{gen} - Words_{human} \rvert$) | **No Judge LLM used:** All evaluation encoders are deterministic, non-Gemini neural and lexical metrics. | Eliminates same-family LLM evaluation bias in RQ2; RM-ANOVA and Friedman tests confirm $p < .001$ on both hybrid and pure SBERT style convergence. |
| RQ3: Policy Compliance Dual-Judge Audit (*step3_rq3_experiment.py*) | Deterministic regex rule verifier (*det_pass*), exact Exemplar Entity Leakage Rate (ELR), and deterministic Slot Grounding $F_1$ | **Primary Judge (*gemini-2.5-flash*) + Independent Validator Judge (*gemini-2.5-pro*):** Cross-verifies natural-language enforcement of 5 enterprise policy rules on stratified audit subsample ($n$ = 48). | Guided *pilot_v1* → *pilot_v2* prompt ordering fix; final dual-judge raw agreement = 95.8%, Cohen's κ = .917 (PABAK = .917), and 93.1% agreement (κ = .861) with deterministic regex verification (Table C3). |

| Research question & pipeline stage | Primary hypothesis-test metrics (Deterministic / Mathematical) | LLM annotator / Judge model & role | Cross-verification agreement & gatekeeping outcome |
|---|---|---|---|
| RQ4: Blind Simulator Persona Fidelity Audit (*step4_rq4_simulation.py*) | Deterministic turn counter to agreed package quotation (*total_turns*, *log_turns*), regex Redundant Elicitation Rate (RER), and goal completion rate | **Independent Blind Validator (*gemini-2.5-pro*):** Reads simulated customer transcripts ($n$ = 45, 15 per archetype) blinded to true labels to verify persona fidelity; contributes zero metrics to RQ4 ANOVA. | Classified 43/45 simulated users into their intended archetype (95.6% accuracy vs. 33.3% chance, Cohen's $\kappa$ = .933, $p < .001$; Table C3), validating the multi-turn simulation environment. |

*Note.* All LLM annotation and judge calls use deterministic decoding (temperature = 0.0, Pydantic JSON schema enforcement) and are cached in *data/interim/* and *data/final/* for 100% offline reproducibility.